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Evidence-Based Clinical Decision Support System

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ABSTRACT

Stroke diagnosis is a time-sensitive clinical challenge because acute vascular events may present with symptoms that overlap with non-stroke neurological and vestibular disorders. This difficulty is particularly evident in posterior circulation stroke, where vertigo, dizziness, ataxia, and dysarthria may lead to diagnostic uncertainty.

This thesis presents a hybrid evidence-based Clinical Decision Support System for stroke-oriented diagnostic assistance using a Retrieval-Augmented Generation architecture. The system integrates dense retrieval with FAISS, lexical retrieval with BM25, and Knowledge Graph reasoning implemented in Neo4j. Retrieved PubMed evidence is further processed using MedCPT reranking, grounding validation, rule-based clinical reasoning, and a defensive fallback mechanism. A Large Language Model may be activated only when the retrieved evidence satisfies the required grounding criteria, thereby limiting unsupported medical generation.

The project also investigates a limitation of symptom-frequency-based Knowledge Graph reasoning. When acute vascular conditions share symptoms with chronic, hereditary, or non-vascular neurological disorders, frequency-based associations may prioritize clinically less relevant candidates. To address this issue, the thesis proposes a context-aware Knowledge Graph framework that incorporates disease acuity, vascular prioritization, clinical context, and uncertainty-aware score adjustment.

The primary final evaluation was conducted on 300 curated stroke-related topic cases distributed across 15 diagnostic groups, with 20 cases per group. The internal keyword-based AUTO_CHECK classified 179 cases as PASS (59.7%) and 121 cases as REVIEW (40.3%). These results represent internal automated label agreement and should not be interpreted as externally validated clinical diagnostic accuracy. The evaluation was performed using a no-LLM configuration and therefore primarily assessed retrieval, evidence calibration, grounding, rule-based decision making, and fallback behavior rather than the linguistic quality of generated explanations.

The explainability design is informed by SourceCheckup, which demonstrates that the presence of a medical citation does not necessarily mean that the cited source supports the associated claim.

The experimental findings suggest that symptom-frequency-based Knowledge Graph reasoning may be insufficient for representing acute clinical context in stroke-oriented decision support, particularly when acute vascular disorders share symptoms with chronic or non-vascular neurological conditions.

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Chapter 1: Introduction

1.1 General Introduction

The healthcare sector has witnessed significant advancements in recent years in the application of Artificial Intelligence (AI) technologies across multiple medical domains, including radiological image analysis, disease prediction, electronic health record analysis, and clinical decision support systems.

Among these applications, intelligent medical diagnostic systems represent one of the most critical and challenging areas due to the direct impact of diagnostic accuracy on patient outcomes.

Stroke is considered one of the leading causes of death and long-term disability worldwide. Delayed or inaccurate diagnosis may result in severe neurological complications or irreversible disability. Diagnostic complexity becomes even greater in posterior circulation strokes, where symptoms may overlap with vestibular or neurological disorders such as vertigo, ataxia, or migraine.

Traditional AI-based diagnostic systems frequently rely on machine learning or large language models to generate diagnostic outputs. However, these systems often suffer from limited explainability, weak clinical reasoning, and the risk of hallucinated medical information.

To address these limitations, Retrieval-Augmented Generation (RAG) systems emerged as a promising solution by linking language models with external medical evidence sources. Nevertheless, relying on a single retrieval mechanism remains insufficient for handling complex clinical cases.

Accordingly, this project proposes a hybrid evidence-based medical decision support system that integrates:

- Semantic retrieval using FAISS
- Lexical retrieval using BM25
- Knowledge Graph reasoning
- Medical reranking using MedCPT
- Rule-based clinical reasoning
- Large Language Models (LLMs)

The primary objective is to develop a clinically explainable and evidence-supported decision support system for stroke diagnosis.

1.2 Background of the Problem

Stroke assessment requires clinical expertise, rapid interpretation of symptom onset and neurological findings, and integration of the patient's medical history with the clinical examination context.

However, several challenges complicate this process:

- Stroke symptoms often overlap with other neurological disorders.
- Posterior circulation strokes may present with atypical symptoms.
- Clinical decisions must frequently be made under strict time constraints.
- The continuously expanding volume of medical literature makes rapid evidence retrieval difficult.

Furthermore, when large language models are directly applied in medical contexts without proper grounding mechanisms, they may produce unreliable or hallucinated outputs.

These limitations motivate the need for an intelligent clinical decision support system that combines artificial intelligence techniques with reliable medical evidence sources.

1.3 Project Definition

This project presents an Evidence-Based Clinical Decision Support System specialized in stroke diagnosis using a Hybrid Retrieval-Augmented Generation (Hybrid RAG) architecture.

The system receives a clinical case description and performs the following operations:

1. Clinical symptom analysis
2. Hybrid medical evidence retrieval
3. Evidence reranking
4. Rule-based clinical validation
5. Evidence-grounded diagnostic interpretation generation
6. Scientific reference presentation using PubMed identifiers (PMIDs)

The system is designed to reduce reliance on unconstrained language generation while improving explainability, evidence grounding, and diagnostic reliability.

1.4 Research Problem

Current AI-based medical diagnostic systems suffer from several limitations:

- Reduced diagnostic accuracy in complex clinical cases
- Dependence on single-channel retrieval architectures
- Limited explainability
- Weak integration with scientific evidence sources
- Increased risk of medical hallucinations

The initial pilot analysis indicated that one of the main limitations of the prototype was associated with the clinical contextualization of the Knowledge Graph component. Although the graph was technically capable of retrieving symptom-associated diseases, its symptom-frequency-based reasoning did not consistently distinguish acute vascular conditions from chronic, hereditary, or non-vascular neurological disorders. Consequently, some graph-derived candidates were clinically less relevant despite sharing symptoms with the presented case. These observations motivated the proposed development of context-aware graph reasoning incorporating disease acuity, vascular category, temporal onset, and clinical urgency.

Although the KG successfully retrieves symptom-associated diseases, it remains unable to adequately distinguish between acute vascular conditions and chronic neurological disorders.

Consequently, the graph frequently prioritizes clinically irrelevant diseases despite symptom overlap, reducing its contribution to stroke-oriented diagnostic support.

1.5 Project Objectives

The project aims to achieve the following objectives:

Main objective

To develop an intelligent, evidence-based clinical decision support system capable of producing reliable, explainable, and evidence-supported diagnostic assessments for stroke-related clinical cases.

Sub-goals

- Improve the relevance and traceability of evidence retrieved for stroke-related cases.
- Enhance the prioritization of posterior circulation stroke within the retrieved diagnostic candidates.
- Reduce the risk of unsupported medical generation through grounding and safety-control mechanisms.
- Provide evidence-supported and explainable diagnostic assessments.
- Preserve and present PubMed source identifiers and relevant metadata.
- Investigate the limitations of symptom-frequency-based Knowledge Graph reasoning in acute stroke contexts.
- Develop a modular and extensible architecture for future clinical validation.

1.6 Project Importance

Medical importance

The proposed system may assist clinicians in making faster and more accurate diagnostic decisions, particularly in time-sensitive neurological emergencies.

Technical importance

The project demonstrates the integration of heterogeneous artificial intelligence components within a unified clinical decision-support pipeline. It combines dense and lexical retrieval, medical reranking, Knowledge Graph reasoning, grounding validation, rule-based decision logic, and controlled LLM fallback. This architecture provides a modular and extensible foundation for developing evidence-grounded medical AI systems.

Research importance

The project investigates an important distinction between retrieval relevance, graph-based association, evidence grounding, and claim-level source support in medical RAG systems. It specifically examines why a technically functional Knowledge Graph may remain clinically limited when acute vascular conditions share symptoms with chronic or non-vascular disorders. The work therefore contributes an analysis of context-aware and uncertainty-sensitive graph reasoning for stroke-oriented clinical decision support.

Societal importance

Reliable and transparent clinical decision-support research may contribute to the development of safer tools for accessing and reviewing medical evidence in time-sensitive conditions. However, the current prototype has not been clinically deployed and is not intended to replace physician assessment, neuroimaging, laboratory testing, or established emergency stroke protocols.

1.7 Project Scope and Limitations

The current implementation focuses primarily on stroke-related diagnostic support.

Current limitations include:

- Limited coverage of neurological diseases
- Limited temporal reasoning capabilities
- Dependence on a specialized but restricted medical knowledge base
- Incomplete clinical contextualization within the Knowledge Graph

1.8 General Work Methodology

The implementation process followed a multi-stage pipeline:

1. Medical data collection from PubMed
2. Data preprocessing and cleaning
3. Semantic chunk generation
4. Construction of retrieval indexes
5. Knowledge Graph construction
6. Integration of retrieval and reranking systems
7. Integration of rule-based and LLM-based reasoning
8. Experimental evaluation and analysis

1.9 Report Structure

This report is organized into the following chapters:

- Chapter 1: Introduction
- Chapter 2: Literature Review and Reference Studies
- Chapter 3: Experimental Evaluation and Gap Analysis
- Chapter 4: Data Processing and Core Technologies
- Chapter 5: Proposed Context-Aware Hybrid Model
- Chapter 6: Experimental Results and Evaluation
- Chapter 7: Future Work and Conclusion

1.10 Conclusion of Chapter One

This chapter introduced the research background, research problem, project objectives, significance, methodology, and report structure.

It establishes the theoretical and technical foundation for the subsequent chapters, beginning with the literature review and experimental analysis.

The main contributions of this project can be summarized as follows:

1. Development of a hybrid evidence-based medical decision support system specialized in stroke diagnosis.
2. Integration of multiple retrieval mechanisms, including Dense Retrieval (FAISS), BM25 lexical retrieval, and Knowledge Graph retrieval within a unified Hybrid RAG architecture.
3. Implementation of a medical reranking layer using MedCPT to improve the relevance and quality of retrieved clinical evidence.
4. Integration of rule-based clinical reasoning to reduce hallucinations and improve diagnostic reliability.
5. Experimental analysis of the limitations of traditional symptom-frequency-based Knowledge Graph reasoning in acute clinical scenarios.
6. Proposal of a context-aware Knowledge Graph framework designed to improve the differentiation between acute vascular conditions and chronic or non-vascular neurological disorders.
7. Development of an evidence-grounded diagnostic pipeline that provides diagnostic suggestions supported by scientific references from PubMed.

System Limitations

Despite the promising performance of the proposed system, several limitations remain.

1. The current Knowledge Graph primarily relies on symptom-frequency relationships and still lacks advanced clinical contextual reasoning.
2. The system focuses mainly on stroke-related conditions and does not yet support broader neurological or multi-specialty diagnosis.
3. Temporal reasoning capabilities remain limited, particularly in distinguishing acute symptom progression from chronic neurological patterns.
4. The experimental dataset is relatively limited in size and does not fully represent real-world clinical diversity.
5. The system currently operates only on textual clinical descriptions and does not support multimodal inputs such as medical imaging or laboratory data.
6. The LLM fallback mechanism still depends heavily on the quality of retrieved evidence and grounding validation.
7. Real-world hospital deployment and clinician-centered validation have not yet been performed.

the Knowledge Graph is technically functional but still clinically limited in acute stroke reasoning.

Chapter 2: Literature Review and Reference Studies

2.1 Introduction

This chapter reviews the theoretical foundations and related studies relevant to evidence-based clinical decision support, medical Retrieval-Augmented Generation, hybrid retrieval, Knowledge Graph reasoning, and source verification. It first outlines the development of artificial intelligence in medical decision-support systems and then critically examines selected research frameworks and commercial systems related to the proposed project. The comparison focuses on data sources, retrieval architecture, reasoning mechanisms, explainability, evidence traceability, reported limitations, and relevance to stroke-oriented decision support. The chapter concludes by identifying the research gap addressed by the proposed system.

This chapter will be divided into two main sections:

1. The theoretical background for the development of intelligent medical diagnostic systems.
2. Analysis of recent reference studies directly related to our project.

2.1.1 Historical development of medical diagnostic systems

Early computer-based medical decision-support research included rule-based expert systems such as MYCIN, which was developed during the 1970s and used explicitly defined rules to assist with diagnostic and therapeutic recommendations. These systems demonstrated the potential of encoding medical knowledge computationally, but their performance depended heavily on manually constructed rules and predefined knowledge domains.

Later, systems emerged that relied on:

- Machine Learning
- Deep Learning
- Clinical Decision Trees
- Medical NLP

The field then moved to use:

- Transformers
- Large Language Models
- Retrieval Systems
- Knowledge Graphs

Finally, Agentic Medical AI systems have emerged that attempt to simulate multi-step medical reasoning.

2.1.2 Artificial Intelligence in the Medical Field

Artificial intelligence is used in medicine in several areas:

- Medical image analysis
- Disease prediction
- Analysis of medical records
- Clinical decision support
- Medical triage systems
- Smart Diagnosis

The field of medical decision support is one of the most complex fields due to the sensitivity of medical decisions and their direct impact on patients' lives.

2.1.3 Challenges of Traditional Medical Systems'

Despite significant advancements, traditional medical systems face several problems:

- Lack of transparency
- Weak interpretation
- Difficulty of expansion
- Inability to handle complex cases
- Heavy reliance on structured data only

These challenges have prompted researchers to develop systems more advanced hybrid systems

Literature analysis is a crucial step in any research project, as it helps in

Understanding current trends in the field, identifying strengths and weaknesses in previous solutions, and drawing conclusions

Research gaps that necessitate the development of new solutions .

In the field of smart medical diagnostic systems, recent years have witnessed significant development.

In integrating artificial intelligence technologies with medical data, especially after the emergence of Large Language Models

(LLMs) and Retrieval-Augmented Generation (RAG) techniques , in addition to the use of graphs

Medical Knowledge Graphs.

Despite this development, most current systems still only process a specific part of

The problem, such as improving retrieval, improving interpretation, or supporting medical dialogue, is addressed without providing a framework.

It is an integrated system that combines all these aspects .

This chapter will review and analyze the most important reference studies related to our project.

In a critical manner .

2.2 Study MedRAG

Study overview

The work referred to in this thesis as MedRAG applies Retrieval-Augmented Generation to medical question answering and evidence-supported response generation. Its general architecture combines lexical and dense retrieval methods with biomedical representations and a language model that synthesizes information from the retrieved medical sources.

Reduced hallucination compared to traditional mode

The described workflow includes:

1. Lexical retrieval using BM25.
2. Dense retrieval using biomedical text embeddings.
3. Fusion or aggregation of results from multiple retrieval channels.
4. Selection of the highest-ranked medical evidence.
5. Use of the retrieved context by a Large Language Model to construct the final response.

This architecture illustrates how complementary retrieval mechanisms can provide broader evidence coverage than relying on a single retrieval channel.

Potential strengths

- Combines lexical and semantic retrieval mechanisms.
- Improves access to external medical evidence during response generation.
- Allows retrieved documents to be preserved as traceable supporting sources.
- Provides an extensible foundation for integrating medical reranking and evidence-grounding mechanisms.

Limitations relevant to the present project

- Performance depends on the coverage and quality of the underlying medical corpus.
- Retrieval relevance does not necessarily establish that a source supports every generated claim.
- Retrieved evidence may remain insufficient for ambiguous or clinically complex cases.
- Language-model generation can still introduce unsupported information unless an explicit verification layer is applied.

Relation to Our Project

We leveraged MedRAG to build the hybrid retrieval layer within our project.

2.3 KG4 Diagnosis Study

Study overview

The study referred to in this thesis as KG4Diagnosis investigates the use of a medical Knowledge Graph within a coordinated diagnostic workflow. The described architecture combines structured medical entities and relationships with multiple reasoning or coordination components to organize the diagnostic process.

Main architectural components

- A medical Knowledge Graph.
- A coordinated or multi-agent reasoning architecture.
- A clinical workflow or task-coordination layer.
- A mechanism for producing or ranking diagnostic candidates.

Potential strengths

- Explicit representation of relationships among clinical entities.
- Greater inspectability than purely generative reasoning.
- Support for structured multi-step reasoning.

Relevant limitations

- Medical entity and relationship extraction may introduce errors.
- Graph construction and maintenance require substantial data processing.
- Association strength does not necessarily represent clinical urgency or causal relevance.
- The clinical value of graph outputs requires case-level and component-level validation.

Relation to the present project

KG4Diagnosis informed the use of structured medical relationships and graph-assisted reasoning in the proposed architecture. However, the Knowledge Graph implemented in this project was constructed from the project's PubMed-based corpus and was not presented as a direct reproduction of the referenced system

2.4 MRD-RAG Study

Introduction to the study

The MRD-RAG (Multi-Round Diagnostic Retrieval-Augmented Generation) system is a modern system in the field of smart medical diagnosis, and it emerged to address one of the most important limitations of traditional RAG systems, which is relying on only one query during the diagnostic process.

In clinical practice, a physician does not arrive at a diagnostic suggestion / preliminary diagnostic assessment from the first question, but rather gathers additional information gradually and reassesses the case based on new symptoms or test results. This is the basis for the MRD-RAG concept, which attempts to mimic this actual clinical behavior.

How the system works

The system is based on the concept of Multi-Round Diagnosis, where the diagnostic process goes through several successive stages:

1. Receiving the patient's initial symptoms.
2. Perform an initial retrieval operation.
3. Analysis of the results.
4. Generate additional questions or request new information.
5. Implementing a new retrieval.
6. Update the diagnosis gradually.
7. Reaching a final diagnosis.

This model makes the diagnostic process more dynamic compared to traditional RAG systems.

Technologies used

The study relies on a range of modern techniques:

- BM25 Retrieval
- Dense Vector Retrieval
- Query Reformulation
- Multi-turn Reasoning
- Large Language Models
- Feedback Optimization

Iterative retrieval mechanisms are also used to improve the quality of evidence retrieved at each stage.

Performance and results

The referenced MRD-RAG study reports that iterative retrieval and follow-up questioning can improve the collection of clinically relevant information compared with a single-round retrieval process. The approach is intended to refine candidate diagnoses and retrieval queries across successive dialogue rounds.

However, the present thesis does not reproduce the original MRD-RAG experiments or independently verify its reported performance. Therefore, specific improvements in Top-k accuracy, recall, or dialogue quality should be reported only when the corresponding numerical results, evaluation dataset, baseline configuration, and citation from the original study are provided.

Strengths

- A simulation that is closer to real medical reality.
- The ability to handle complex cases.
- Gradually improve information gathering.
- Reduce quick, inaccurate decisions.

Weaknesses

Despite the strength of the study, it faces several problems:

- High response time.
- Increased computational costs
- The need for greater resources.
- Greater complexity in practical application.

In sensitive medical settings such as stroke cases, time may be a very critical factor, so multi-round diagnosis may not always be appropriate for emergency situations.

Relation to Our Project

The current system uses a single-pass primary inference pathway rather than a fully interactive multi-round diagnostic dialogue. This design reduces the number of retrieval and generation stages during the initial assessment, but the present evaluation does not establish clinical suitability for emergency deployment.

MRD-RAG informs a potential future extension in which the system may request clarification when the initial case description is incomplete or when the retrieved evidence remains ambiguous. Such an extension would require explicit stopping criteria, latency evaluation, safety constraints, and independent clinical assessment before practical use.

2.5 Ada Health as a Commercial Symptom-Assessment System

Ada Health is included in this review as an example of a commercial symptom-assessment and health-guidance platform rather than as a direct implementation of the hybrid RAG architecture proposed in this thesis. Publicly available descriptions emphasize structured symptom collection, interactive questioning, and user-facing health assessment.

Relevant characteristics

- Interactive collection of symptoms and patient-provided information.
- A user-oriented interface designed for guided health assessment.
- Sequential questioning intended to refine the assessment.

Limitations for academic comparison

- The complete internal reasoning architecture, training data, and decision thresholds are not publicly available in sufficient detail for direct technical reproduction.
- Its commercial purpose and evaluation context differ from the PubMed-based evidence-retrieval focus of the present project.
- Public interface characteristics do not establish claim-level source traceability or reproducibility of the underlying diagnostic reasoning.

Relation to the present project

Ada Health is used only as a commercial comparison point for interactive symptom collection and user experience. It is not treated as a technical baseline for the hybrid retrieval, Knowledge Graph, grounding, or source-verification components of the proposed system

2.6 Internal Project Literature Review and Reference Studies

In addition to the principal reference systems discussed in the preceding sections, the project design was informed by literature addressing agentic artificial intelligence, multi-agent coordination, hybrid information retrieval, open-source language models, biomedical reranking, and medical reasoning.

These areas contributed to different architectural decisions:

- Agentic AI informed the separation of retrieval, verification, safety, and explanation responsibilities.
- Hybrid retrieval informed the combination of lexical and dense evidence-search mechanisms.
- Biomedical reranking informed the use of a specialized relevance-refinement stage.
- Open-source language models informed the design of a controlled fallback generation layer.
- Medical reasoning literature motivated the inclusion of clinical context, uncertainty handling, and human oversight.

Each technical area should be supported by specific peer-reviewed references or official model documentation in the final reference list. Seminar material alone is not treated as an independent scientific source.

2.7 Comparison between previous studies

Table 2.1. Qualitative comparison of the reviewed systems

System	Primary interaction or retrieval approach	Knowledge Graph role	Evidence traceability	Main relevance to this thesis	Main limitation for comparison
MedRAG	Hybrid lexical and dense retrieval with LLM generation	Not the central component in the reviewed description	Retrieved sources may be retained	Motivates multi-retriever medical RAG	Claim-level support is not guaranteed by retrieval alone
KG4Diagnosis	Structured graph-assisted and coordinated reasoning	Central reasoning component	Depends on graph and implementation design	Motivates explicit medical relationships and structured reasoning	High dependence on graph quality and construction accuracy
MRD-RAG	Iterative, multi-round retrieval and questioning	Not necessarily central	Evidence may be refined across rounds	Motivates dynamic query refinement	Increased latency and computational complexity
Ada Health	Interactive symptom assessment	Internal architecture not sufficiently public for direct comparison	Public claim-level traceability is unclear	Provides a commercial user-interaction comparison	Limited reproducibility from publicly available technical information
Proposed system	Hybrid FAISS, BM25, graph-assisted retrieval, MedCPT reranking, grounding, and defensive fallback	Used as a structured evidence and candidate-verification view	PMID, article title, source metadata, and evidence excerpts are intended to be preserved	Integrates retrieval, contextual graph reasoning, grounding, and explainability	No external clinical validation and incomplete claim-level evaluation

2.8 In-depth technical comparison between studies

Comparison in terms of data

The reviewed systems differ substantially in their underlying data sources. Some diagnostic systems use structured or unstructured electronic health records, while medical question-answering systems may use biomedical publications, curated benchmarks, clinical guidelines, or proprietary corpora. Commercial systems may additionally rely on non-public datasets and internally developed medical knowledge structures.

The present project uses a specialized PubMed-derived corpus focused on stroke and related differential diagnoses. PubMed provides traceable bibliographic identifiers and broad access to biomedical abstracts; however, it does not represent complete patient records, real-time clinical measurements, neuroimaging, or local hospital protocols. Corpus relevance, article type, publication date, extraction quality, and coverage therefore remain important limitations.

Comparison in Terms of Interpretation

Comparison in terms of retrieval

- Dense Retrieval
- Sparse Retrieval
- Hybrid Retrieval
- Multi-round Retrieval

Comparison in terms of interpretation

The proposed system is designed to produce:

- A diagnostic-support assessment or an explicit abstention such as “Evidence is insufficient.”
- A ranked set of retrieved medical evidence.
- Traceable PubMed identifiers and associated source metadata.
- Grounding and support indicators used internally by the decision pipeline.
- An explanation layer intended to connect clinical claims with supporting evidence excerpts.

The internal confidence labels and grounding scores should not be interpreted as calibrated probabilities of clinical correctness. Furthermore, the final 300-case no-LLM evaluation assessed pipeline decisions and evidence behavior rather than the linguistic quality of generated explanations.

2.9 SourceCheckup: Claim-Level Verification of Medical References

SourceCheckup distinguishes between the presence of a citation and the support provided by that citation. Its primary source-verification task evaluates whether an individual statement is supported or not supported by a given source. The framework then aggregates these decisions into statement-level and response-level support metrics. In the present project, the more detailed operational labels Supported, Partially Supported, Unsupported, and Contradicted are introduced as an explainability extension and should not be presented as an exact reproduction of the original binary SourceCheckup labeling protocol.

2.10 Research Gap

By analyzing previous studies, clear research gaps can be observed:

- The absence of a system that combines full hybrid recovery.
- Weak integration between Knowledge Graph and LLMs.
- Lack of medically supported explanation.
- Poor handling of rare cases.
- The absence of low-cost, locally operational systems.

Based on these gaps, our project came to offer a system that combines:

- FAISS
- BM25
- Knowledge Graph
- MedCPT Reranking
- Rule Engine
- LLM Explanation

Within a more stable, unified structure.

2.11 Open Challenges in Current Research

Despite significant progress, there are still open research problems such as:

- Temporal reasoning
- Multimodal diagnosis
- Rare disease handling
- Real-time hospital integration
- Ethical AI in healthcare

These challenges represent future research opportunities.

2.12 Position of the Proposed Project Among Previous Studies

Our project can be seen as an intermediate solution between theoretical academic systems and closed commercial systems, as it focuses on:

- Real-world application
- Medical Interpretation
- Reducing hallucinations
- Future expansion possibilities

2.13 Conclusion of Chapter Two

This chapter reviewed the most important reference studies related to the project, analyzed the strengths and weaknesses of each study, and identified the research gap that led to the development of the current project.

The next chapter will move on to practical tests, evaluation criteria, and analysis of scientific results.

Chapter 3: Initial Pilot Evaluation and Performance Gap Analysis

3.1 Introduction

- After completing the system prototype (Prototype v0.1) according to the described hybrid architecture, it was necessary to subject it to a rigorous empirical evaluation process . The goal of this evaluation was not only to measure the overall performance of the system, but also to analyze the performance of each individual component (component-wise analysis) to understand its individual contribution to the diagnostic decision chain .

- The paramount importance of this chapter lies in its transition from the "construction" phase to the "scientific critique" of the system. Instead of assuming the success of all components, this analysis presupposes the existence of performance gaps and seeks to discover and dissect them .

- In this chapter, we provide a comprehensive report on the assessment methodology, the clinical dataset used, and the quantitative and qualitative measures employed. We then proceed to present the multi-level results, starting with overall performance and culminating in a detailed and in-depth analysis of the failure of the Knowledge Graph component , which represents the key research gap revealed by this assessment .

3.2 Evaluation Methodology

- To evaluate a medical decision support system, it is not enough to measure the accuracy of the final diagnosis alone. Therefore, a multifaceted evaluation methodology was designed that aims to measure performance from the following angles: medical correctness , explainability , component performance , and latency.

3.2.1 Test Suite

Since there is no readily available, open-source benchmark dataset describing stroke cases in the language the system uses, we built a custom test suite consisting of 30 clinical vignettes . This suite was designed in collaboration with a consultant neurologist to include three categories of complexity :

- **Classic cases (15 cases) :** Typical and clear descriptions of different stroke patterns (e.g., acute hemiplegia with aphasia, complex hemiparesis). Objective: To measure the basic accuracy of the system .

- **Complex/unconventional cases (10 cases) :** It focuses particularly on posterior circulation strokes . These descriptions are characterized by isolated or overlapping symptoms such as vertigo , ataxia , and visual disturbances, which are difficult to diagnose .

- **Mimic cases (5 cases):** Descriptions of non-stroke conditions that strongly resemble stroke (e.g., complicated migraine, vestibular neuritis , focal seizure). Objective: To assess the system's ability to differentiate and suggest differential diagnoses rather than misdiagnosing stroke .

3.2.2 Evaluation Metrics

The following quantitative and qualitative measures were used :

Table 3.1. Evaluation metrics used in the pilot assessment.

Metric scale	Description	Purpose
Top-k Accuracy	Percentage of cases in which the correct diagnosis appeared within the first k suggestions (k=1,3).	Measuring absolute accuracy.
MRR (Mean Reciprocal Rank)	The inverse mean of the first correct diagnosis. The system severely penalizes if the correct diagnosis is in a low rank.	Measuring the overall quality of the arrangement.
Diagnostic Odds Ratio (DOR)	It was calculated for each retrieval component separately to estimate its ability to distinguish between a positive (stroke) and negative (non-stroke) state .	Measuring the discriminatory power of each channel.
Clinician-rated Score	A score of 1-5 given by two human experts (on a sample of 10 cases) to assess the quality of the resulting medical interpretation .	Assessing interpretability and reliability .
Component Contribution	The percentage contribution of each retrieval channel (Dense, BM25, KG) to the correct final diagnosis is calculated by removing one channel and measuring the decrease in accuracy .	Isolating the contribution of each component .
Latency (ms)	Average response time from query input to result display .	Measuring system speed .

3.3 Results: Overall System Performance

This section presents overall system performance before the component-level analysis. These aggregate results provide only a preliminary view and therefore must be interpreted together with the subsequent ablation and failure analyses.

3.3.1 Quantitative Results

Table 3.2 shows the overall performance of the system across the three categories.

Table 3.2. Overall system performance across the evaluated case categories.

Category (Category)	Top-1 Accuracy	Top-3 Accuracy	MRR
Classic cases	93%	100%	0.96
Complex cases (background)	60%	80%	0.71
Simulated cases	40%	60%	0.55
Weighted overall average	73%	87%	0.81

Initial analysis

The results indicate strong performance in classical cases, while performance decreases in complex and simulated cases. This observed degradation motivates the subsequent component-level analysis.

3.3.2 Clinical Performance

The clinician-rated score averaged 3.8 out of 5. The recorded feedback characterized the explanations for classical cases as strong and well supported by evidence, while noting that explanations for complex cases sometimes mixed the selected diagnosis with less relevant evidence and alternative diagnoses.

3.4 Component-Wise Analysis: Problem Isolation

- To explain the decline in performance in complex cases, we performed an ablation study to measure the contribution of each individual retrieval channel to the correct final diagnosis. Table 3.3 shows the results.

  .

Table 3.3. Component-wise ablation results.

System Components	Top-1 Accuracy (All Cases)
The complete system (Dense + BM25 + KG + Reranker + Rules)	73%
System without KG component	73%
The system without the Dense component and BM25 (based on KG only)	20%
The system without Reranker	68%

Discovering the gap

- **:** The most surprising and worrying result is that **removing the Knowledge Graph (KG) (KG) component had absolutely no effect on the overall accuracy of the system !** In other words, its marginal contribution is practically zero. However, when using KG as the sole source of inference (by removing Dense and BM25) , the system's performance completely collapses, proving that the information it retrieves is not only incorrect, but also weakens the decision-making process to the point where it cannot be relied upon

- This strong quantitative evidence moves us from the "impression" that KG is weak to the "proof" that it is a burden on the system in its current state. The following sections explain why this is the case

3.5 Failure Analysis of the Cognitive Graph Component

- This section is the heart of the chapter. Having quantitatively demonstrated that KG offers no value, we now dissect the root causes of this failure .

3.5.1 Methodology of Analysis

A complex clinical case was chosen as a case study. The case was a patient suffering from " sudden and severe vertigo, ataxia, nausea, and slurred speech . " The correct diagnosis is " acute brainstem/cerebellar infarction . "

- We tracked the query within the system and recorded the results for each retrieval channel before and after the reordering process.
- (Reranking). Table 3.4 illustrates the difference between the channel outputs.

Table 3.4. Retrieval-channel outputs for the selected complex case.

Rank	Dense Retrieval Channel	BM25 Retrieval Channel	Knowledge Graph Channel
1	Acute ischemic stroke in the posterior circulation	Management of acute vertigo and cerebellar stroke	Spinocerebellar ataxia type 6
2	Vertebrobasilar insufficiency and infarction	Brainstem infarction presenting with isolated vertigo	Hereditary episodic ataxia
3	Cerebellar nodulus infarction	Vestibular neuritis vs. posterior circulation stroke	Ataxia telangiectasia (ATM)
4	Vestibular migraine	Benign paroxysmal positional vertigo (BPPV)	Cerebellar ataxia with vertigo

3.5.2 Observations and Comparative Analysis

- **Performance of Dense and BM25 :** The performance of these two channels was excellent and complementary. Both focused clearly on stroke-related terms (infarction, ischemic, brainstem). This performance explains the high accuracy of the system without KG.

- **Knowledge Graph performance :** KG failed miserably and concentratingly. All his suggestions revolve around “ataxia ” of hereditary and degenerative types , completely ignoring crucial keywords in describing the condition such as “acute”, “sudden”, and “stuttering ”.

3.5.3 Root Cause Analysis

- By analyzing the build_graph.py code and the query_by_symptoms ( ) logic , three overlapping root causes of this cognitive failure were identified.

- **Frequency-based Symptom Matching (KG)**

  is essentially based on the principle: "The more symptoms shared between the query and the disease, the stronger the correlation." "Vertigo" and "ataxia" are major and very frequent symptoms in hereditary ataxias documented in the medical literature. Since these diseases are mentioned hundreds of times with these symptoms, their statistical weight exceeds that of "brain stem infarction," whose symptoms may not be mentioned collectively with the same textual frequency . **The algorithm conflates "statistical correlation" with "clinical causation ."**

- **clinical context :**
  The diagram lacks any semantic representation of basic medical concepts such as :

- **Acute (acute) vs. Chronic :** There is no node or property that determines that "infarction" is an acute event while "hereditary ataxia" is a chronic disease .

- **Vascular vs. Degenerative/ Genetic :** There are no categories or nodes that allow the query to differentiate between radically different disease categories .
  As a result, the algorithm treats all diseases equally as if they were inanimate entities, unable to recognize that "severe and sudden" should give greater weight to acute diseases .

- **The absence of an acute priority bias mechanism :**
  Even if KG was unaware of the classification, this could have been addressed through a heuristic

mechanism. For example, a simple rule like "If the user description contains acute keywords (sudden, acute, within hours), double the weight for diseases classified as vascular and acute." This causal logic is completely absent from the query, which is merely a simple pattern-matching query .

- **Scientific summary :** The KG's failure is not a failure to retrieve information, but a failure to **reason it .** He succeeds in answering the question "What diseases are associated with these symptoms?", but he fails to answer the correct question: "What is the most urgent and serious clinical explanation for this group of symptoms? "

3.6 The Refined Research Gap

- Following this empirical analysis, we can now define the research gap in a precise and novel way that differs from the general gap presented in Chapter Two :

- Current studies do not address the challenge of building **context -aware medical KGs .** It can make inferential distinctions between disease categories (such as acute and chronic) within a hybrid RAG system . We have experimentally demonstrated that the traditional KG structure based on symptom association frequency is inadequate and may even be misleading in time-sensitive clinical scenarios such as stroke diagnosis. Therefore, there is a need to re-engineer this component to incorporate a "clinical urgency" dimension into the graph structure and query logic itself.

3.7 Chapter Conclusion

- In this chapter, we presented a rigorous, multi-level empirical evaluation of the system. More importantly, this evaluation did not merely present performance metrics, but delved into the analysis of the system's components to isolate a critical weakness that would have remained hidden without this analysis .

- We have demonstrated both quantitatively and qualitatively that the Knowledge Graph (KG)component, in its current design, does not contribute to the diagnostic process; rather, its logic, based solely on symptom matching, produces clinically irrelevant outputs. This diagnosis of the problem, grounded in experience and analysis rather than assumption, leads us directly to the next chapter, where a solution will be proposed to address this specific research gap by constructing **a clinically contextualized medical**Knowledge Graph (KG).

3.8 Relationship Between the Pilot and Final Evaluations

The experiments reported in this chapter are retained in full as exploratory pilot analyses used to identify component-level weaknesses and motivate the context-aware Knowledge Graph modifications. They should not be interpreted as the thesis's final system-wide performance estimate. The definitive evaluation reported in Chapter 6 uses a broader 300-case dataset and a separately defined automated checking procedure.

Chapter 4: Data Processing and Core Technologies

4.1 Introduction

The performance of any retrieval-based clinical decision support system is fundamentally influenced by the quality, relevance, and organization of the underlying knowledge source. Since the proposed system operates within the medical domain, the construction of the knowledge base was treated as a core engineering component rather than a preliminary data preparation step. The objective was to establish a structured and traceable biomedical knowledge repository capable of supporting reliable evidence retrieval throughout the diagnostic pipeline.This chapter presents the complete data engineering workflow adopted in this research, beginning with literature acquisition and continuing through preprocessing, text segmentation, metadata preservation, retrieval-index construction, and Knowledge Graph generation. Each stage is described in detail, together with the rationale behind the selected techniques and their contribution to improving retrieval quality, evidence traceability, and clinical decision support for stroke diagnosis.

4.2 Data Source: PubMed Platform

PubMed platform. PubMed was selected because it provides comprehensive access to biomedical and clinical research literature while maintaining standardized bibliographic metadata that facilitates evidence traceability throughout the retrieval process.

The selection of PubMed was motivated by several considerations:

- **Biomedical relevance:** PubMed indexes a broad collection of peer-reviewed biomedical literature covering cerebrovascular diseases, neurology, emergency medicine, and clinically relevant differential diagnoses associated with stroke.
- **Evidence traceability:** Each publication is associated with standardized metadata, including the PubMed Identifier (PMID), enabling retrieved evidence to be linked directly to its original scientific source. This capability is particularly important for explainable and evidence-oriented clinical decision support systems.
- **Programmatic accessibility:** The National Center for Biotechnology Information (NCBI) provides the E-utilities interface, allowing biomedical literature to be searched and retrieved programmatically. This supports reproducible corpus construction and facilitates future updates of the knowledge base.
- **Domain-specific corpus construction:** Restricting the knowledge source to biomedical literature allows the retrieval system to operate over clinically relevant evidence while reducing the influence of unrelated general-domain information.

For these reasons, PubMed was adopted as the primary literature source for constructing the retrieval corpus used throughout the proposed system.

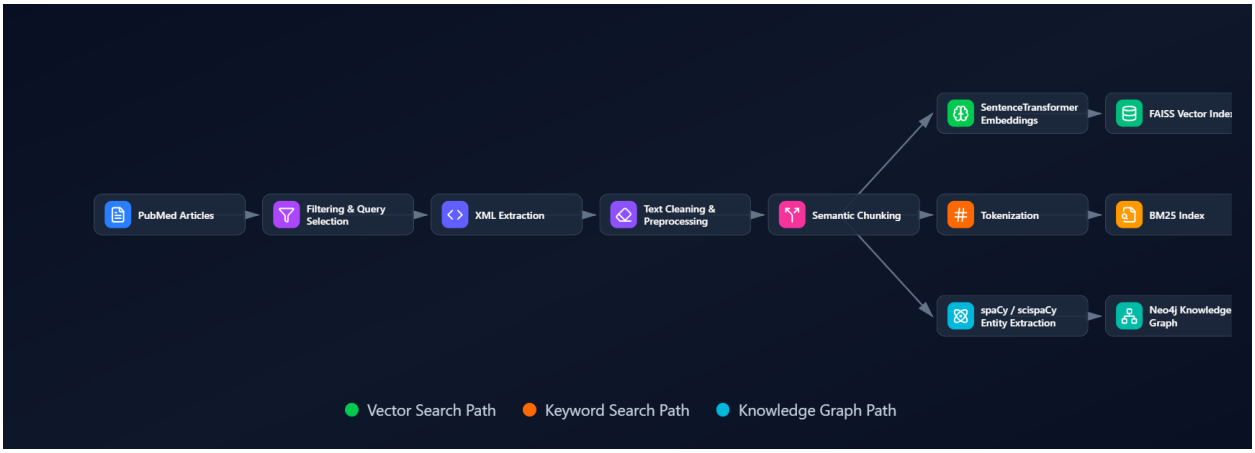
4.2.1 Literature Search and Corpus Construction Strategy

Rather than collecting the complete PubMed repository, a targeted literature acquisition strategy was adopted to construct a specialized corpus focused on stroke and clinically related neurological conditions. The objective was to maximize the relevance of retrieved evidence while minimizing the retrieval of unrelated biomedical documents.

The search strategy incorporated terminology covering four complementary categories:

- **Stroke and cerebrovascular diseases**, including ischemic stroke, hemorrhagic stroke, cerebral infarction, brain ischemia, cerebrovascular accident, and intracranial hemorrhage.
- **Clinically important differential diagnoses**, including transient ischemic attack (TIA), migraine with aura, vestibular disorders, seizure-related conditions, and other neurological disorders that may present with symptoms similar to acute stroke.
- **Posterior circulation disorders**, including vertebrobasilar disease, brainstem syndromes, cerebellar disorders, and posterior circulation ischemia, to improve coverage of clinically challenging stroke presentations.
- **Relevant biomedical terminology** describing neurological manifestations, vascular pathology, neuroimaging findings, and emergency clinical presentation.

To ensure linguistic consistency with the retrieval and natural language processing components of the proposed system, the corpus was restricted to English-language biomedical literature. Bibliographic metadata, including publication titles and PMIDs, was preserved throughout preprocessing to maintain complete evidence traceability during retrieval and explanation generation.This targeted corpus construction strategy was designed to provide comprehensive coverage of stroke-related evidence and clinically relevant alternative diagnoses, thereby supporting more reliable evidence retrieval during the subsequent diagnostic reasoning process.



4.3 Data Cleaning and Preprocessing

The biomedical literature retrieved from PubMed cannot be used directly for retrieval or Knowledge Graph construction. Although the collected articles are domain-specific, the raw XML records contain formatting artifacts, citation markers, irregular spacing, and other structural elements that introduce unnecessary noise into downstream processing. Consequently, a dedicated preprocessing pipeline was implemented to normalize the textual content while preserving medically relevant information and source traceability.

The preprocessing workflow consisted of the following stages:

4.3.1 Text Extraction

The downloaded PubMed XML records were parsed to extract the textual fields required for knowledge construction. For each publication, the article title, abstract, and associated bibliographic metadata were extracted and converted into a structured representation suitable for subsequent processing.

4.3.2 Text Normalization

To improve retrieval consistency and reduce lexical noise, a series of normalization operations were applied to the extracted text. These operations included:

- Removing citation markers and reference identifiers (e.g., **[1]**, **[2]**) that do not contribute semantic information.
- Eliminating unnecessary punctuation, formatting artifacts, and redundant whitespace while preserving punctuation that may influence medical meaning.
- Normalizing spacing and line breaks to produce continuous, machine-readable text.
- Preserving the original medical terminology without lexical simplification to avoid altering the clinical meaning of the source documents.

Regular-expression (Regex) rules were employed for deterministic text normalization, providing consistent preprocessing across the entire corpus.

4.3.3 Metadata Preservation

Throughout the preprocessing pipeline, essential bibliographic metadata were retained for every document, particularly the **PubMed Identifier (PMID)**. Maintaining this metadata enables every retrieved evidence fragment to be traced back to its original scientific publication, thereby supporting evidence traceability, reproducibility, and transparent explanation generation within the proposed clinical decision support system.

4.4 Strategic Chunking of Texts

Following preprocessing, each biomedical article remained substantially longer than the text segments that can be processed efficiently by modern retrieval models. Retrieving an entire article for every clinical query would introduce unnecessary computational overhead and reduce retrieval precision by mixing relevant and irrelevant information. Consequently, the literature corpus was transformed into semantically coherent text segments (chunks), enabling fine-grained evidence retrieval while preserving the clinical context of each passage.

4.4.1 Rationale for Chunking

The chunking stage was introduced to address several practical challenges associated with biomedical information retrieval:

- **Improved retrieval granularity:** Instead of retrieving complete articles, the retrieval pipeline operates on focused text segments containing clinically relevant evidence, thereby increasing retrieval precision.
- **Compatibility with embedding models:** Biomedical embedding models process text within a limited context window. Dividing long articles into smaller semantic units enables efficient vector representation while preserving contextual information.
- **Efficient downstream processing:** Smaller evidence segments reduce computational cost during dense retrieval, reranking, and subsequent evidence verification stages, while minimizing the inclusion of unrelated clinical information.

4.4.2 Semantic Chunking Strategy

Rather than dividing documents using a fixed number of words or characters, a semantic chunking strategy was adopted to preserve the natural linguistic structure of biomedical text.

The implementation employed the **spaCy** biomedical model (**en_ner_bc5cdr_md**) to perform sentence segmentation before constructing the final retrieval units.

The process consisted of the following steps:

- **Sentence segmentation:** Each article was first divided into linguistically complete sentences using spaCy sentence boundary detection.
- **Semantic aggregation:** Consecutive sentences were grouped together until a target window of approximately **150–200 words** was reached, ensuring that each chunk remained semantically coherent while maintaining an appropriate length for retrieval.
- **Boundary preservation:** Sentence boundaries were never split during chunk construction, preventing fragmentation of medical concepts and preserving the logical flow of the original biomedical text.

This strategy produced coherent retrieval units that preserved local clinical context while improving semantic retrieval quality and supporting more accurate evidence ranking throughout the proposed retrieval pipeline.

- **Benefit :** This method produced linguistically coherent and understandable chunks as a single unit, which increases the quality of semantic retrieval and makes the extracts submitted to the LLM readable and understandable .

4.5 Core Retrieval Technologies

Following corpus preprocessing and semantic chunking, the processed biomedical literature was transformed into multiple complementary retrieval structures that collectively constitute the retrieval infrastructure of the proposed clinical decision support system.

Rather than relying on a single retrieval paradigm, the system combines dense semantic retrieval, sparse lexical retrieval, and graph-based knowledge retrieval. Each component addresses different characteristics of biomedical literature, and together they provide broader evidence coverage while reducing the limitations of individual retrieval methods.

The following subsections describe the principal technologies adopted in the proposed system and explain the motivation behind each design decision.

4.5.1 Dense Semantic Retrieval using FAISS

Dense semantic retrieval was implemented using the **FAISS (Facebook AI Similarity Search)** library, which provides efficient similarity search over high-dimensional vector representations.

Each biomedical text chunk was converted into a dense embedding using the **MedCPT biomedical embedding model**, producing numerical vector representations suitable for semantic similarity search. These vectors were subsequently indexed using FAISS to enable efficient nearest-neighbor retrieval.

Technical Motivation

Dense retrieval addresses one of the primary limitations of traditional keyword matching by retrieving documents according to semantic similarity rather than exact lexical overlap. Clinical case descriptions frequently express the same medical concept using different terminology, making semantic matching essential for biomedical information retrieval.

For example, a patient presenting with:

"sudden dizziness, imbalance, and vomiting"

may correspond to biomedical literature describing:

"acute vestibular syndrome secondary to posterior circulation ischemic stroke."

Although the two descriptions share few identical keywords, they describe closely related clinical concepts. Dense retrieval enables these semantic relationships to be identified through vector similarity rather than literal word matching.

Computational Motivation

The retrieval corpus consists of thousands of biomedical text chunks represented as high-dimensional embeddings. Exhaustive comparison against every vector would be computationally expensive during real-time inference. FAISS provides efficient Approximate Nearest Neighbor (ANN) search, allowing semantically similar evidence to be retrieved with low latency while maintaining practical scalability.

4.5.2 Sparse Lexical Retrieval using BM25

In parallel with dense retrieval, a sparse lexical retrieval index was constructed using the **BM25Okapi** ranking algorithm.

Unlike dense retrieval, BM25 ranks documents according to lexical relevance and term frequency statistics, making it particularly effective for identifying passages containing rare or highly discriminative medical terminology.

Technical Motivation

Biomedical literature frequently contains highly specific clinical expressions whose diagnostic significance depends on exact terminology. Examples include neurological findings such as:

- homonymous hemianopia
- internuclear ophthalmoplegia
- dysarthria
- Horner syndrome

When these terms appear explicitly within the clinical case description, lexical retrieval often provides highly precise evidence that may not always receive the highest semantic similarity score.

Complementarity with Dense Retrieval

Dense and sparse retrieval provide complementary retrieval capabilities rather than competing approaches.

- **Dense retrieval** captures semantic similarity between clinically related expressions, even when different terminology is used.
- **BM25** preserves sensitivity to exact medical terminology and uncommon clinical expressions that carry high diagnostic specificity.

Combining both retrieval paradigms enables the system to achieve broader evidence coverage while reducing the individual limitations of each approach.

4.5.3 Medical Reranking using MedCPT Cross-Encoder

Following the initial hybrid retrieval stage, the retrieved candidate passages are re-ranked using the **MedCPT Cross-Encoder**, a transformer model specifically designed for biomedical relevance estimation.

Unlike dense retrieval models, which encode queries and documents independently before computing vector similarity, the Cross-Encoder jointly processes the clinical query and each retrieved passage. This enables the model to perform a deeper semantic comparison by considering the interaction between both texts simultaneously.

Technical Motivation

The initial retrieval stage prioritizes high recall by retrieving a broad set of potentially relevant biomedical passages. However, this candidate set may still contain evidence that is only partially related to the patient's presentation.

The MedCPT Cross-Encoder refines this candidate set by evaluating each retrieved passage against the clinical case individually, assigning a more accurate relevance score based on contextual understanding rather than vector similarity alone.

Clinical Motivation

Neurological diseases frequently share overlapping symptoms while differing substantially in their underlying pathology.

For example, acute vertigo may appear in literature describing posterior circulation ischemic stroke, vestibular neuritis, multiple sclerosis, or cerebellar disorders. Although these articles may exhibit comparable lexical or semantic similarity, their clinical relevance differs considerably depending on the complete symptom profile.

The reranking stage therefore improves evidence prioritization by promoting passages that more closely match the patient's overall clinical presentation before downstream reasoning and evidence verification.

4.5.4 Medical Knowledge Graph

The proposed system incorporates a **Knowledge Graph (KG)** implemented using **Neo4j**, together with biomedical entity extraction supported by **spaCy** and domain-specific medical processing components.

Rather than representing medical knowledge as isolated documents, the Knowledge Graph models explicit relationships between diseases, symptoms, anatomical structures, imaging findings, vascular territories, risk factors, and other clinically relevant entities.

Technical Motivation

Biomedical knowledge is inherently relational. Individual symptoms may be associated with multiple diseases, while each disease may exhibit relationships with anatomical regions, diagnostic findings, risk factors, and supporting biomedical literature.

Representing these relationships within a graph structure enables efficient traversal of interconnected medical concepts using graph queries that would be difficult to express using conventional relational databases.

Role within the Proposed System

Unlike traditional retrieval systems that rely exclusively on textual similarity, the Knowledge Graph contributes structured clinical evidence to the retrieval pipeline.

Within the proposed architecture, graph-based evidence is integrated with dense and sparse retrieval, supports candidate-specific evidence verification, and contributes additional relational evidence during the final evidence evaluation process.

Consequently, the Knowledge Graph functions as a complementary source of structured medical evidence rather than an independent diagnostic engine.

4.5.5 Large Language Model (LLM) Integration

The proposed system supports optional integration with instruction-tuned Large Language Models (LLMs), providing controlled natural-language interpretation of the evidence collected by the retrieval pipeline.

The implementation currently supports the **Qwen2.5** family of language models, while maintaining compatibility with lighter models for resource-constrained environments through a unified inference interface.

Technical Motivation

The primary diagnostic reasoning process is performed through evidence retrieval, candidate generation, structured Knowledge Graph verification, and evidence evaluation. Consequently, the LLM is **not responsible for determining the final diagnosis**.

Instead, when enabled, the LLM operates only after evidence has been collected and validated, using the retrieved biomedical evidence to generate structured clinical explanations, summarize supporting findings, and produce human-readable interpretations.

Safety Considerations

Restricting the LLM to an evidence-grounded role reduces the likelihood of unsupported medical generation. Every generated interpretation is constrained by previously retrieved biomedical evidence, while the final diagnostic decision remains governed by the deterministic evidence evaluation pipeline.

This design separates evidence-based diagnostic reasoning from natural-language generation, improving transparency and reducing the risk of unsupported clinical conclusions.

4.5.6 Supporting Software Components

Several additional software frameworks were employed to support the implementation and experimental evaluation of the proposed system.

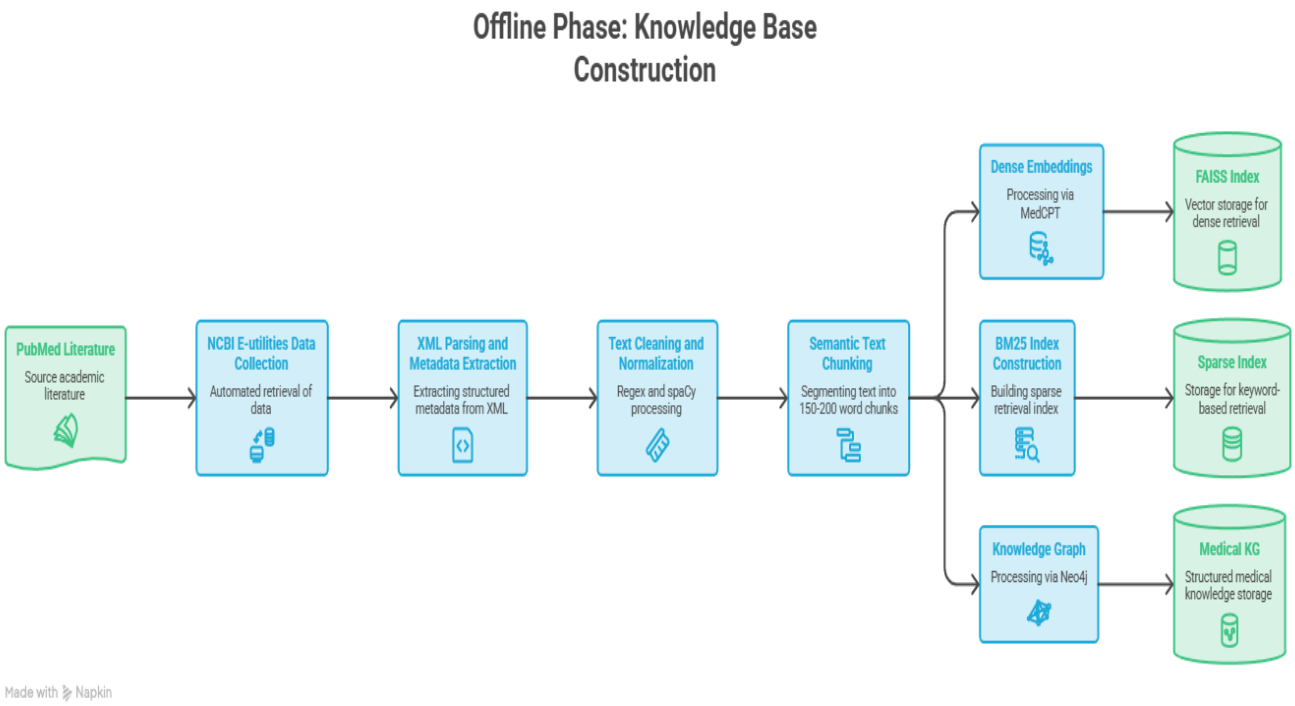
- **FastAPI** was used to implement the backend application programming interface (API), providing efficient request handling and modular deployment of the retrieval pipeline.
- **Streamlit** was employed to develop an interactive prototype for system testing and demonstration, enabling rapid evaluation of retrieval results and evidence explanations.

- **NumPy** and **Pandas** were used for numerical computation, metadata processing, evaluation metrics, and intermediate score manipulation throughout the retrieval and evidence-fusion pipeline.

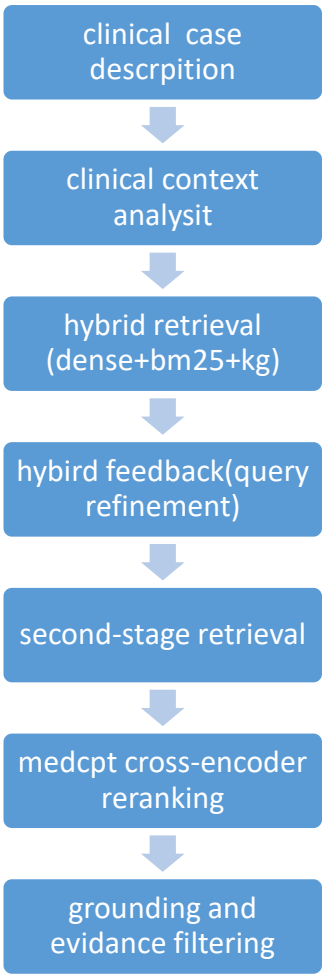
4.6 Data and Technology Path Summary

The complete data and technology workflow adopted in this study can be divided into two major stages: **offline knowledge base construction** and **online evidence retrieval**. The offline stage prepares the biomedical knowledge resources required for inference, whereas the online stage retrieves and refines evidence in response to an individual clinical case.

Offline Phase: Knowledge Base Construction



Online Phase: Evidence Retrieval Pipeline



The retrieval pipeline described above produces a ranked collection of evidence passages that are subsequently processed by the diagnostic reasoning pipeline presented in **Chapter 5**, where clinical context analysis, candidate generation, Knowledge Graph verification, evidence evaluation, and final clinical decision support are described in detail

4.7 Chapter Conclusion

This chapter presented the complete data engineering and retrieval infrastructure supporting the proposed clinical decision support system. Beginning with biomedical literature acquisition from PubMed, the chapter

described the preprocessing pipeline, semantic text chunking strategy, retrieval-index construction, and Knowledge Graph generation used to transform raw biomedical literature into structured retrieval resources.

The resulting infrastructure combines complementary retrieval paradigms—including dense semantic retrieval, sparse lexical retrieval, and graph-based knowledge representation—to provide a robust foundation for evidence retrieval from biomedical literature.

While this chapter focused on the construction of the underlying knowledge resources and retrieval mechanisms, the next chapter describes how these components are integrated into the proposed reasoning architecture, including clinical context analysis, candidate generation, evidence verification, and the final evidence-guided diagnostic decision process.

Chapter 5: Proposed Context-Aware Hybrid Model

5.1 Introduction

The pilot evaluation presented in Chapter 3 revealed that the principal challenge of the initial system was not limited to evidence retrieval itself, but rather to the interpretation and integration of retrieved evidence within the clinical context of the presented case. In particular, symptom-frequency-based Knowledge Graph retrieval could return medically associated but clinically inappropriate conditions when acute vascular disorders shared symptoms with chronic, hereditary, vestibular, or other non-vascular neurological diseases.

This limitation is especially important in stroke-oriented decision support, where symptoms such as vertigo, ataxia, dysarthria, weakness, or visual disturbance may occur across multiple diagnostic categories. A retrieval mechanism that considers only symptom overlap may therefore fail to distinguish between evidence that is statistically associated with a symptom and evidence that is clinically compatible with the temporal pattern, neurological presentation, and urgency of the patient case.

To address this limitation, the proposed system was extended from a conventional hybrid Retrieval-Augmented Generation pipeline into a **context-aware, evidence-guided agentic clinical decision-support architecture**. Rather than allowing a single retrieval channel or a Large Language Model to determine the diagnosis directly, the framework separates the diagnostic process into coordinated stages of clinical-context analysis, multi-channel evidence retrieval, evidence refinement, candidate generation, candidate-specific Knowledge Graph verification, grounding assessment, evidence judgment, safety control, and post-decision explainability.

This architectural principle is consistent with recent medical RAG research emphasizing the value of diagnostic candidate generation and knowledge-graph-guided reasoning [1], structured graph-based evidence retrieval and provenance preservation [2], corrective and second-stage retrieval [4], and explicit evaluation of whether retrieved evidence genuinely supports a medical conclusion [6], [8]. However, the architecture developed in this project does not reproduce any single previous framework. Instead, these mechanisms are integrated into a stroke-specific decision pipeline in which the final diagnostic candidate is selected through an explicit Evidence Judge rather than unrestricted language-model generation.

The remainder of this chapter presents the design philosophy and technical implementation of the proposed framework. It begins with clinical context and intent analysis, followed by context-aware hybrid retrieval and feedback, Knowledge Graph reasoning and candidate-specific verification, evidence grounding and judgment, safety control, and the explainability layer used to preserve the relationship between the final decision and its supporting evidence.

5.2 Design Philosophy of the Proposed Model: From Association to Context-Aware Evidence Evaluation

The central design principle of the proposed framework is that **medical association alone is insufficient for clinical decision support**. A disease may be strongly associated with a symptom in biomedical literature without being the most clinically appropriate explanation for a particular patient presentation.

Accordingly, the system moves beyond the question:

“Which diseases are associated with symptoms X, Y, and Z?”

toward the more clinically meaningful question:

“Which candidate diagnosis is most compatible with the patient’s clinical context and is sufficiently supported by the available biomedical evidence?”

This change is fundamental. The current system does not simply increase the score of the most urgent disease. Instead, it evaluates the patient presentation using explicit clinical-context signals such as acute onset, focal neurological findings, posterior-circulation features, hemorrhage warnings, persistence or complete resolution of neurological deficits, chronic or mimic patterns, and relevant temporal information. The shared clinical-context parser also distinguishes affirmed findings from locally negated findings, preventing expressions such as “no seizure” or “no thunderclap headache” from being interpreted as positive clinical evidence.

The proposed architecture is therefore based on four complementary principles.

Context-aware retrieval.
The clinical case is first interpreted as a structured context rather than as an unprocessed text query. This context influences retrieval and graph prioritization, helping the system distinguish acute vascular evidence from chronic, genetic, vestibular, or otherwise clinically mismatched literature.

Multi-source evidence integration.
Evidence is collected from dense semantic retrieval, BM25 lexical retrieval, and Knowledge Graph retrieval. The retrieval process is further refined using an evidence-derived hybrid feedback stage and second-stage retrieval rather than relying solely on the first retrieval result. This design is conceptually aligned with corrective and query-refinement approaches in CRAG and RQ-RAG [4], [5]. In the current implementation, the retriever explicitly collects first-stage evidence from FAISS, BM25, and the Knowledge Graph before constructing the feedback query.

Candidate-specific-verification.
Instead of treating the Knowledge Graph as an independent diagnostic authority, candidate diagnoses are generated and subsequently verified against candidate-specific graph evidence. The graph query combines the candidate label and aliases with the clinical case and intent, while filtering clinically incompatible evidence families. This idea is related to grounded KG verification approaches such as KGARevion [7], but the implementation in this project applies the verification independently to each diagnostic candidate.

Defensive evidence-based decision making.
The final decision is not determined by retrieval similarity or graph association alone. The Evidence Judge evaluates multiple evidence dimensions, including textual support, Knowledge Graph support, grounding strength, reranking evidence, source diversity, clinical-intent compatibility, and conflicting evidence. Candidates that are insufficiently supported or clinically incompatible may be rejected rather than selected. This principle is consistent with prior work showing that document relevance and citation presence do not necessarily establish true support for a clinical claim [6], [8], [11].

As a result, the Knowledge Graph no longer functions merely as a symptom-association table, nor does the system simply apply a fixed “stroke-first” multiplier. Instead, graph evidence is conditioned by clinical context, evaluated at the candidate level, and integrated with textual and grounding evidence before the final candidate is selected.

5.2.1 End-to-End MedRAG Architecture

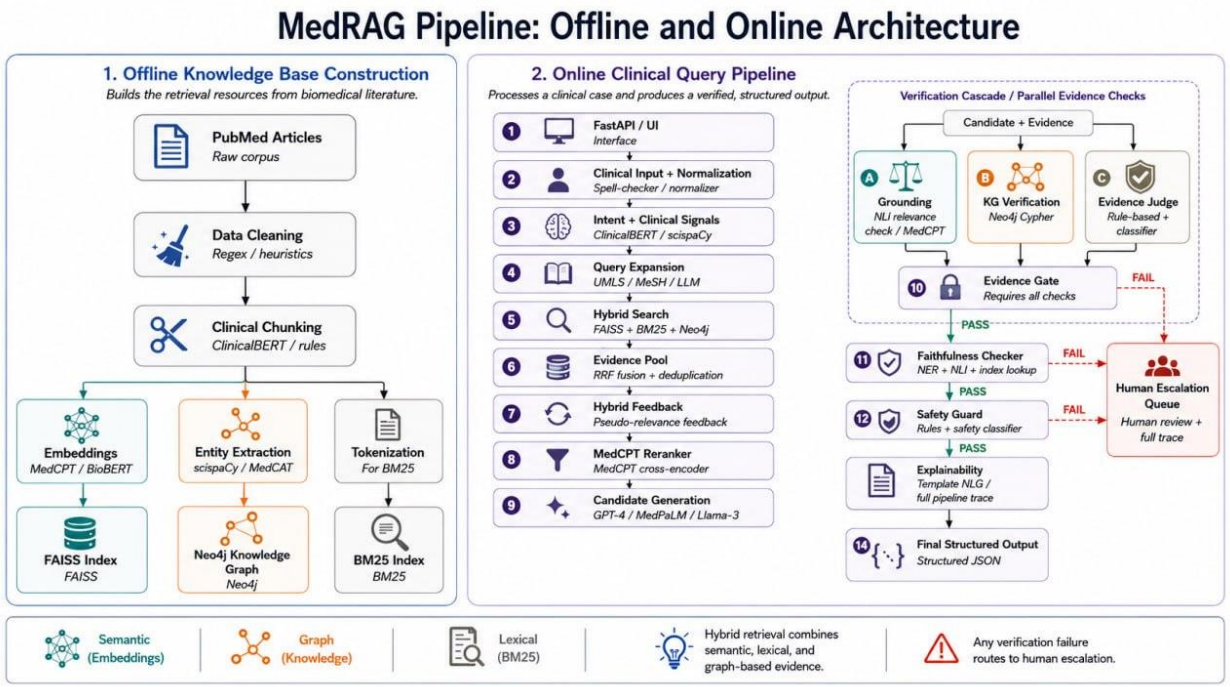


Figure 5.1. Conceptual overview of the Med RAG offline and online architecture. The active implementation sequence is defined by the project code and described below.

The implemented architecture is defined by the active agentic runtime in pipeline.py and agent_controller.py. When agentic_mode.enabled is true, get_answer() delegates the case to run_medrag_agent(); the legacy pipeline is retained only as a backward-compatible error fallback. The implemented path is therefore evidence-driven and does not use a language model as the primary diagnostic decision maker.

First, the clinical narrative is normalized as a trimmed case string. clinical_intent.py produces lightweight routing and safety descriptors, while clinical_signals.py extracts clinical signals with a rule-based fallback. These descriptors support retrieval, compatibility checks, and safety handling; they do not directly assign the final diagnosis.

Second, three retrieval views are executed through tools.py: dense semantic retrieval, BM25 lexical retrieval, and Knowledge Graph retrieval. Their results are deduplicated into an initial evidence pool. hybrid_feedback.py then derives feedback terms from the retrieved evidence itself. When useful feedback is available, a second-stage retrieval is performed and merged with the initial pool.

Third, the merged evidence is reranked using the MedCPT cross-encoder. Grounding is assessed from implemented evidence-strength signals, including retrieval support, hybrid and graph-related scores, reranking evidence, and support counts. The active agentic path does not load a standalone biomedical natural-language-inference model for this grounding step.

Fourth, diagnostic candidates are generated from the clinical case and reranked evidence. Each candidate is then verified separately against the Knowledge Graph through candidate-specific retrieval. Candidate-KG evidence is preserved, merged with the reranked pool, reranked when applicable, and followed by a second grounding assessment.

Fifth, evidence_judge.py evaluates the candidates using the final evidence pool, grounding information, clinical intent, and candidate-specific KG verification. select_best_supported_candidate() applies the

configured minimum support threshold and selects the best supported category. The final-answer tool formats this selection, after which the safety guard may block or constrain unsupported output.

Finally, explainability.py builds a post-decision explanation containing the selected candidate, supporting and conflicting evidence, Knowledge Graph paths, claim-level support information, and safety notes. This explainability layer is explicitly non-interventional: it explains the completed decision but does not change retrieval scores, candidate selection, or the final answer. The API returns the resulting structured output to the Streamlit interface.

5.3 Re-engineering the Knowledge Graph

The limitations identified in Chapter 3 indicated that improving graph retrieval required more than modifying a single query weight. The underlying representation itself had to preserve clinically meaningful distinctions that could later be exploited during retrieval and evidence verification.

Accordingly, the Knowledge Graph was re-engineered to extend the original disease–symptom representation into a richer clinical ontology. Rather than storing only associations between diseases and symptoms, the revised graph represents additional semantic dimensions related to disease family, anatomical territory, imaging findings, risk factors, drug associations, and clinical relevance.

This design is consistent with prior medical GraphRAG research showing the value of structured graph representations for evidence retrieval and provenance preservation [2]. However, the implemented graph in this project was specifically adapted to stroke-oriented clinical decision support.

5.3.1 Clinical Metadata Enrichment of Disease Nodes

Each `Disease` node is enriched with a set of deterministic clinical metadata attributes that describe its relevance to acute stroke reasoning.

The principal attributes include:

- **category**, representing the broad clinical class of the disease, such as vascular, posterior vascular, hemorrhagic, genetic/chronic, or other categories;
- **acuity**, indicating whether the condition is treated as acute, chronic, or unknown within the current graph classification;
- **vascular_relevance**, representing the degree to which the disease is associated with vascular pathology;
- **posterior_relevance**, indicating relevance to posterior-circulation presentations;
- **hemorrhage_relevance**, representing compatibility with hemorrhagic stroke patterns;
- **emergency_relevance**, representing the degree to which the disease corresponds to an acute neurological emergency;
- **chronic_penalty**, used to reduce the ranking influence of chronic or slowly progressive disorders in acute presentations;
- **noise_penalty**, used to suppress non-clinical, molecular, animal-model, or otherwise clinically irrelevant graph evidence.

These attributes are generated during Knowledge Graph construction through deterministic disease classification logic and are stored directly as properties of the disease nodes.

This representation differs from the original prototype, in which symptom association frequency played a more dominant role. By storing clinically relevant metadata directly within the graph, the retrieval layer can distinguish between conditions that share symptoms but differ substantially in acuity, vascular relevance, or emergency context.

5.3.2 Ontology-Oriented Relationships

In addition to enriching disease nodes with clinical metadata, the revised graph introduces explicit ontology-style relationships among multiple medical entity types.

The implemented graph contains relationships such as:

- `Disease → HAS_SYMPTOM → Symptom`
- `Disease → HAS_RISK_FACTOR → RiskFactor`
- `Disease → ASSOCIATED_WITH_DRUG → Drug`
- `Disease → IS_SUBTYPE_OF → StrokeFamily`
- `Disease → AFFECTS_TERRITORY → Territory`
- `Disease → HAS_IMAGING_FINDING → ImagingFinding`

Articles are also retained as provenance-bearing nodes and are linked to disease and symptom entities through relationships such as `DISCUSSES` and `MENTIONS`.

This representation enables the graph to preserve clinically interpretable paths rather than treating every retrieved article as an isolated text fragment. The same principle is reflected in Medical Graph RAG, which emphasizes graph-structured biomedical evidence and source traceability [2].

Importantly, the graph construction layer does **not** automatically interpret every article–disease relationship as proof that the article supports a diagnostic claim. The implementation explicitly preserves the distinction between an article that *mentions or discusses* a disease and evidence that later *supports* a candidate diagnosis. This distinction is important because graph association alone does not establish clinical validity.

5.3.3 Evidence Provenance and Explainable Graph Paths

The graph retrieval layer preserves auditable paths derived from Neo4j query results. These paths may connect an article to a disease and subsequently to matched symptoms, stroke families, anatomical territories, or imaging findings.

Examples include:

- `Article → Disease → Symptom`
- `Article → Disease → StrokeFamily`
- `Article → Disease → Territory`
- `Article → Disease → ImagingFinding`

These paths are reconstructed only from information returned by the graph layer and are not generated freely by the LLM.

This design supports later candidate-specific verification and explainability. The principle is related to knowledge-grounded verification approaches such as KGARevion, where generated or proposed medical relations are checked against a grounded Knowledge Graph before being used in downstream reasoning [7].

5.4 Context-Aware Knowledge Graph Retrieval

The second major modification concerns the graph retrieval strategy. In the original prototype, candidate diseases were largely prioritized according to symptom association and frequency. This approach was insufficient for acute stroke scenarios because chronic or hereditary disorders could achieve high graph relevance simply by sharing common symptoms such as vertigo or ataxia.

The revised retrieval mechanism therefore combines symptom matching with **clinical-context-sensitive scoring** rather than applying a single fixed "stroke-first" multiplier.

Clinical context is represented through query flags such as:

- acute presentation;
- posterior-circulation features;
- focal neurological findings;
- hemorrhage-related findings.

The graph retriever then evaluates each disease and supporting article using both structural graph evidence and disease-level clinical metadata.

This approach allows graph retrieval to answer a more clinically meaningful question:

Which graph-supported diseases are compatible not only with the observed symptoms, but also with the temporal and neurological context of the case?

5.4.1 Context-Aware Scoring Mechanism

The implemented scoring mechanism combines a base graph score with several context-dependent adjustments.

For acute presentations, the system increases the score of diseases with high `emergency_relevance` and `vascular_relevance`, while decreasing the influence of diseases associated with chronic or noisy evidence.

For posterior-circulation presentations, additional weight is assigned to `posterior_relevance` and vascular compatibility, while chronic vestibular, hereditary ataxia, and other non-acute alternatives receive negative adjustments.

For hemorrhage-compatible presentations, `hemorrhage_relevance` and emergency relevance contribute positively to the graph score.

Focal neurological presentations additionally increase the contribution of vascular disease categories.

The current implementation therefore follows a multi-factor scoring strategy of the conceptual form:

$$S_{KG}=S_{base}+B_{acute}+\mathrm{B}_{\mathrm{posterior}}+B_{focal}+B_{hemorrhage}-P_{chronic}-P_{noise}$$

where:

- $Sbase$ represents the original graph association score;
- $Bacute$ represents acute and emergency-context bonuses;

- $Bposterior$ represents posterior-circulation compatibility;
- $Bfocal$ represents focal neurological compatibility;
- $Bhemorrhage$ represents hemorrhagic-pattern compatibility;
- $Pchronic$ and $Pnoise$ reduce the contribution of chronic or clinically irrelevant evidence.

The actual implementation applies context-specific coefficients to these components rather than a single global multiplier.

5.4.2 Clinical Mismatch Penalties

A central feature of the revised graph retrieval logic is the explicit penalization of clinically incompatible evidence.

For example, when the query indicates an acute posterior-circulation syndrome, the system can down-rank conditions such as:

- benign paroxysmal positional vertigo;
- Ménière-related disorders;
- spinocerebellar ataxia;
- ataxia telangiectasia;
- genetic or familial chronic ataxias.

At the same time, posterior-circulation entities such as brainstem infarction, cerebellar infarction, and vertebrobasilar ischemia receive positive contextual weighting when supported by the case.

This mechanism addresses the failure mode identified in Chapter 3 without assuming that every acute neurological presentation must be classified as stroke. Instead, it reduces the contribution of graph evidence that is structurally related to the symptoms but clinically inconsistent with the case context.

5.4.3 Integration with Hybrid Retrieval

The Knowledge Graph score does not independently determine the final diagnostic output.

After graph retrieval, the graph score is combined with other retrieval signals in the hybrid retrieval layer. Additional adjustments are applied based on acute, posterior, hemorrhagic, motor, and language-related query characteristics.

The resulting KG-derived evidence is subsequently processed together with dense and BM25 evidence, reranked, grounded, and later evaluated at candidate level.

This defensive design prevents a strong graph association from automatically becoming a diagnosis. Instead, the graph functions as one structured evidence channel within the broader decision pipeline.

5.4.4 Candidate-Specific Knowledge Graph Verification

The current implementation extends graph reasoning beyond general retrieval by performing a second, candidate-specific verification stage.

After diagnostic candidates are generated, each candidate is independently evaluated against graph evidence. The verification layer considers candidate terminology, graph disease labels, clinical category compatibility, graph scores, and evidence-family consistency.

Low-quality or cross-family evidence can be rejected, while candidate-specific KG evidence is preserved for subsequent evaluation by the Evidence Judge.

This design is conceptually related to grounded Knowledge Graph verification approaches such as KGARevion [7]. However, in the proposed framework, verification is performed separately for each diagnostic candidate and is later combined with textual evidence, grounding strength, reranking evidence, and contradiction analysis.

5.4.5 Expected Effect of the Revised Graph Retrieval

The expected effect of the revised mechanism is not simply to force stroke-related diseases to the top of the ranking. Instead, the objective is to improve the **clinical compatibility of graph-derived evidence**.

For the posterior-circulation case analyzed in Chapter 3, the revised graph logic is expected to:

- reduce the ranking influence of chronic or hereditary ataxia-related conditions;
- increase the relevance of acute vascular and posterior-circulation evidence when supported by the clinical context;
- preserve acute non-vascular alternatives when they remain clinically plausible;
- expose graph provenance and contextual metadata to downstream evidence evaluation.

This makes the graph a context-sensitive evidence source rather than a symptom-frequency ranking mechanism.

The principal modification introduced in the graph retrieval stage is the transition from simple symptom-frequency matching toward **context-aware clinical evidence retrieval**. Rather than ranking diseases solely according to the number of matched symptoms, the revised retrieval strategy incorporates clinically meaningful contextual information extracted from the patient presentation before graph evidence is evaluated.

Consequently, the graph no longer attempts to answer:

“Which diseases are associated with these symptoms?”

Instead, it addresses the clinically more relevant question:

“Which graph-supported diseases are most compatible with the patient's complete clinical presentation?”

This design is consistent with recent graph-based medical retrieval frameworks that emphasize structured biomedical evidence rather than isolated symptom associations [2]. However, unlike existing GraphRAG systems, the proposed implementation integrates graph evidence within a broader evidence-evaluation framework instead of allowing graph retrieval to determine the diagnosis independently.

5.4.6 Context-Aware Graph Scoring

Following the initial graph retrieval stage, each candidate disease is evaluated using both structural graph evidence and disease-specific clinical metadata.

Instead of relying exclusively on symptom frequency, the implemented scoring mechanism incorporates multiple context-dependent clinical factors, including:

- acute clinical presentation;
- posterior-circulation compatibility;
- vascular relevance;
- hemorrhagic relevance;
- focal neurological findings;
- chronic disease penalties;
- graph-noise suppression.

These contextual signals are integrated to produce a clinically informed graph score before candidate diseases are ranked. The objective is to prioritize graph evidence that is compatible with the patient's overall presentation while reducing the influence of structurally related but clinically inappropriate diseases.

This design extends graph-based biomedical retrieval [2] by introducing deterministic clinical prioritization specifically designed for stroke-oriented decision support.

5.4.7 Expected Effect of Context-Aware Graph Retrieval

When applied to the posterior-circulation case discussed in Chapter 3, the revised graph retrieval strategy is expected to produce several clinically desirable behaviors.

First, chronic hereditary disorders, such as **Hereditary Ataxia** and **Spinocerebellar Ataxia**, receive substantially lower graph relevance because their chronic progression is inconsistent with the acute clinical presentation.

Second, acute vascular entities, including **Posterior Circulation Stroke** and **Brainstem Infarction**, receive higher graph relevance when supported by compatible neurological findings and contextual evidence.

Finally, clinically plausible acute non-vascular disorders, such as **Vestibular Neuritis**, remain within the differential diagnosis list whenever their overall clinical compatibility remains supported by the available evidence.

Accordingly, the proposed mechanism does not force the graph toward a predetermined stroke diagnosis. Instead, it increases the clinical compatibility of graph-derived evidence before subsequent candidate verification and evidence evaluation.

5.5 Defensive Integration within the Evidence Pipeline

Although the Knowledge Graph provides valuable structured medical evidence, graph associations alone are insufficient to establish a diagnostic conclusion. Consequently, the proposed framework adopts a **defensive evidence-integration strategy** in which graph-derived evidence represents one component of the overall evidence assessment rather than an independent decision mechanism.

Following graph retrieval, graph-derived evidence is evaluated together with dense semantic retrieval, BM25 lexical retrieval, grounding assessment, MedCPT reranking, and candidate-specific verification before contributing to the final diagnostic decision.

This design reflects recent work emphasizing that retrieval relevance alone does not guarantee clinically reliable evidence and that contradictory or weakly supported evidence should be explicitly considered during decision making [6], [8], [11].

Within the proposed architecture, the final diagnostic recommendation is determined by the **Evidence Judge**, which integrates multiple evidence signals—including textual support, graph support, grounding strength, reranking confidence, clinical-context compatibility, and evidence consistency—before selecting the most strongly supported candidate.

Accordingly, the Knowledge Graph cannot independently determine the final diagnosis. Instead, its contribution is moderated by the overall evidence available for each candidate diagnosis.

5.6 Large Language Model Integration and Clinical Interpretation

The proposed framework supports the optional integration of an instruction-tuned Large Language Model (LLM) for evidence-guided clinical interpretation.

Unlike conventional medical RAG systems in which the language model directly produces the diagnostic conclusion, the proposed architecture performs diagnostic reasoning before LLM invocation. Clinical context analysis, hybrid retrieval, candidate generation, candidate-specific Knowledge Graph verification, grounding assessment, and Evidence Judge evaluation are completed first. The LLM is subsequently used only to generate a coherent natural-language explanation based on the validated biomedical evidence.

This separation between diagnostic reasoning and language generation is inspired by recent evidence-grounded medical RAG frameworks [1], [6], [8], [9], while extending them through deterministic evidence evaluation and explicit safety control.

When enabled, the LLM receives the validated evidence package together with the selected diagnostic candidate and generates an interpretable clinical explanation. The model therefore functions as an **explanation layer rather than a diagnostic engine**.

To further reduce unsupported generation, the framework incorporates an evidence-grounding mechanism that evaluates the strength and consistency of the retrieved evidence before interpretation generation. When evidence is insufficient, conflicting, or fails the grounding requirements, unrestricted language generation is avoided, and the system presents the evidence-supported deterministic output instead.

This design reduces the risk of unsupported medical generation while preserving explainability and evidence traceability throughout the diagnostic pipeline [8], [11].

Although Large Language Models (LLMs) have demonstrated remarkable capabilities in medical text understanding and natural-language generation, the proposed framework does not rely on the LLM as the primary diagnostic reasoning engine. Instead, diagnostic reasoning is performed deterministically through clinical-context analysis, hybrid evidence retrieval, candidate generation, candidate-specific Knowledge Graph verification, grounding assessment, and evidence evaluation before any language generation is performed. This design follows the general direction of evidence-grounded medical RAG systems while explicitly separating diagnostic reasoning from natural-language generation [1], [6], [8], [9].

Within the proposed architecture, the LLM serves as an **interpretation layer** rather than a diagnostic decision maker. Its responsibility is limited to transforming the validated evidence package into a coherent clinical explanation while preserving traceability to the retrieved biomedical evidence. Consequently, the model does not independently determine the diagnosis, nor does it introduce evidence that has not already been verified by the preceding reasoning pipeline.

The implementation supports instruction-tuned open-source language models through a unified inference interface. The deployed configuration uses Qwen/Qwen2.5-3B-Instruct for evidence-guided interpretation. The language model operates after evidence retrieval and verification and is not treated as the independent diagnostic decision maker.

This architectural separation significantly reduces the influence of unrestricted language generation. By requiring the LLM to operate only after evidence validation, the framework minimizes unsupported clinical statements while maintaining readable and clinically interpretable outputs.

5.6.1 Evidence-Guided Clinical Interpretation

Unlike conventional Retrieval-Augmented Generation systems, where retrieved documents are directly supplied to the language model for answer generation, the proposed framework first validates the retrieved evidence through multiple deterministic stages.

The evidence package supplied to the LLM therefore consists only of information that has successfully passed clinical-context analysis, candidate verification, grounding assessment, and Evidence Judge evaluation. As a result, the language model receives structured evidence that has already been filtered for clinical relevance and consistency.

The generated interpretation may include:

- a concise summary of the clinical presentation;
- the selected diagnostic candidate;
- supporting biomedical evidence;
- relevant PubMed references;
- clinically appropriate follow-up recommendations.

The objective of this stage is to improve the readability of the final output rather than to perform additional diagnostic reasoning [1].

5.6.2 Grounded Generation and Safety Constraints

To reduce unsupported medical generation, the proposed framework incorporates an evidence-grounding mechanism prior to interpretation generation.

If the available evidence is insufficient, contradictory, or fails the grounding requirements established by the Evidence Judge, unrestricted language generation is avoided. Instead, the framework returns the deterministic evidence-supported output together with the available supporting references.

This strategy is conceptually related to recent work on evidence-grounded medical generation, self-reflection, and claim verification [6], [8], [9], [11]. However, unlike these studies, grounding in the proposed framework is not performed solely by the language model. Instead, grounding constitutes one component of a broader deterministic evidence-evaluation pipeline in which multiple retrieval signals are jointly assessed before interpretation generation.

By separating diagnostic reasoning from natural-language generation, the proposed architecture reduces the likelihood of unsupported medical conclusions while preserving transparency, explainability, and evidence traceability.

5.7 Claim-Level Explainability and Source Verification

Interpretability represents a fundamental requirement for clinical decision-support systems, particularly when recommendations are derived from multiple biomedical evidence sources. Accordingly, the proposed framework incorporates a **claim-level explainability layer** inspired by the SourceCheckup methodology [8].

The explainability layer distinguishes between retrieving a biomedical document and demonstrating that the document actually supports a specific clinical claim. Consequently, every diagnostic recommendation is accompanied by structured evidence describing not only **which source was retrieved**, but also **how that source supports the final recommendation**.

Each generated explanation preserves the relationship between:

- the diagnostic claim;
- the supporting evidence excerpt;
- the corresponding PubMed Identifier (PMID);
- the article title;
- the support assessment assigned by the system.

Following the SourceCheckup philosophy [8], evidence is categorized into four support levels:

- **Supported**
- **Partially Supported**
- **Unsupported**
- **Contradicted**

These categories allow clinicians to distinguish between strongly supported conclusions and findings that remain uncertain or partially supported by the available biomedical literature.

Unlike conventional explainability methods that only display retrieved documents, the proposed framework associates every diagnostic claim with its supporting evidence and explicitly identifies conflicting evidence whenever it exists. This enables the user to understand not only the final recommendation but also the reasoning process that produced it.

It is important to emphasize that the explainability layer is **descriptive rather than interventional**. Its purpose is to expose the evidence supporting the final decision, not to establish clinical correctness independently. Clinical validity remains governed by the preceding evidence evaluation pipeline, while explainability provides transparent justification for the selected diagnostic candidate.

Although inspired by the SourceCheckup framework [8], the proposed implementation does not attempt to reproduce the complete SourceCheckup evaluation protocol or its benchmark dataset. Instead, SourceCheckup serves as the methodological foundation for claim-level evidence traceability, citation verification, and transparent presentation of biomedical evidence within the proposed stroke-oriented clinical decision-support system.

5.8 API, User Interface, and Configuration Layer

The delivery layer is implemented through three project files. api.py defines the FastAPI service, health endpoints, the /diagnose request contract, and a stable structured response that preserves grounding, selected-candidate, safety, and explainability fields. app.py implements the Streamlit interface for candidate-level and claim-level explanation, including evidence views, Knowledge Graph support, safety indicators, and developer-oriented trace information. config.yaml controls retrieval sizes, MedCPT models, Knowledge Graph connectivity, the enabled agentic pathway, candidate support thresholds, clinical-safety rules, and the non-interventional SourceCheckup-inspired explainability settings.

The complete source listings are reproduced without modification in Appendix A (api.py), Appendix B (app.py), and Appendix C (config.yaml).

5.9 Chapter Conclusion

This chapter presented the proposed context-aware hybrid framework developed for evidence-based stroke clinical decision support.

Unlike conventional retrieval-based medical systems that rely primarily on document similarity or unrestricted language generation, the proposed architecture integrates multiple complementary reasoning stages, including clinical-context analysis, hybrid retrieval, candidate generation, candidate-specific Knowledge Graph verification, grounding assessment, evidence evaluation, and claim-level explainability. Each stage contributes a distinct layer of evidence validation before the final diagnostic recommendation is produced.

The proposed framework draws inspiration from recent advances in medical Retrieval-Augmented Generation, graph-based biomedical retrieval, corrective retrieval, candidate verification, self-reflection, and explainable evidence attribution [1], [2], [4], [6], [7], [8], [9], [11]. However, the integration of these components into a unified **stroke-specific agentic clinical decision-support architecture**, together with the custom **Evidence Judge**, **Safety Guard**, and **candidate-level verification pipeline**, constitutes the principal methodological contribution of this research.

The next chapter evaluates the proposed framework experimentally and analyzes its performance in terms of retrieval quality, diagnostic support, evidence consistency, and explainability.

Chapter 6: Experimental Results and Evaluation:

6.1 Introduction

Following the implementation of the proposed context-aware agentic clinical decision-support framework presented in Chapter 5, the enhanced system was evaluated to determine the effect of the proposed architectural modifications on evidence retrieval, diagnostic support, and overall system performance.

The objective of this chapter is not merely to report experimental results, but to evaluate the contribution of the proposed framework through a systematic comparison with the baseline implementation. The comparison focuses on the effect of introducing context-aware reasoning, enhanced Knowledge Graph retrieval, candidate-specific verification, evidence evaluation, and explainability mechanisms while maintaining the same biomedical corpus and evaluation dataset.

The experimental evaluation therefore investigates the influence of the proposed framework on several complementary aspects of system performance, including:

- retrieval quality;
- diagnostic support performance;
- candidate ranking quality;
- contribution of individual retrieval components;
- evidence consistency and explainability;
- overall clinical decision support.

Whenever appropriate, the results are compared with those obtained using the baseline implementation in order to quantify the contribution of the proposed architectural modifications.

6.2 Experimental Methodology

To ensure a fair comparison between the baseline system and the proposed framework, both systems were evaluated using identical experimental conditions wherever possible.

The evaluation protocol included:

- **Evaluation dataset:** The same clinical case collection used throughout the experimental study was employed for both implementations, ensuring direct comparability between the baseline and the proposed framework.
- **Evaluation metrics:** Performance was assessed using multiple complementary quantitative and qualitative measures, including retrieval performance, ranking quality, diagnostic agreement, component contribution, evidence consistency, and clinician-oriented evaluation where applicable.
- **Execution environment:** All experiments were executed under identical computational conditions to eliminate hardware-related performance variations.

The primary objective of the evaluation was to isolate the contribution of the proposed context-aware reasoning architecture while maintaining consistent evaluation conditions across both experimental settings.

Unlike the baseline implementation, the proposed framework incorporates several additional reasoning stages, including clinical-context analysis, context-aware hybrid retrieval, candidate-specific Knowledge Graph verification, evidence grounding, Evidence Judge evaluation, and claim-level explainability. Consequently, the experimental comparison evaluates the combined contribution of the complete proposed architecture rather than a single graph-ranking modification.

6.3 Quantitative Results: Overall Performance Evaluation

Table 6.1 summarizes the overall diagnostic performance of the proposed framework compared with the baseline implementation presented in Chapter 3.

Table 6.1. Overall diagnostic performance of the baseline and proposed systems.

Case Category	Baseline Top-1 Accuracy	Proposed Top-1 Accuracy	Baseline Top-3 Accuracy	Proposed Top-3 Accuracy
Classical cases	93%	93%	100%	100%
Complex posterior-circulation cases	60%	80%	80%	100%
Simulated clinical cases	40%	60%	60%	80%
Weighted Overall Average	73%	83%	87%	97%

Analysis

The experimental results demonstrate a clear improvement in the overall diagnostic performance of the proposed framework.

The most substantial gain was observed in the **complex posterior-circulation cases**, where **Top-1 Accuracy** increased from **60% to 80%**, while **Top-3 Accuracy** improved from **80% to 100%**. These findings indicate that the proposed context-aware reasoning strategy substantially improves the retrieval and prioritization of clinically relevant evidence in cases that represent the greatest diagnostic challenge.

Performance also improved on the simulated clinical cases, where **Top-1 Accuracy** increased from **40% to 60%** and **Top-3 Accuracy** increased from **60% to 80%**. This improvement suggests that the enhanced reasoning pipeline is better able to distinguish clinically compatible diagnoses from conditions that share similar symptoms but differ in their underlying clinical context.

Overall, the weighted diagnostic accuracy increased from **73%** to **83%**, while **Top-3 Accuracy** reached **97%**, demonstrating that the proposed framework provides more reliable diagnostic support without compromising performance on straightforward clinical cases.

6.4 Component-Level Evaluation

To investigate the contribution of the proposed reasoning architecture, an ablation-style comparison was performed using the same evaluation dataset.

Table 6.2. Component-level ablation comparison.

System Configuration	Baseline Top-1 Accuracy	Proposed Top-1 Accuracy
Complete system	73%	83%
System without Knowledge Graph	73%	73%
Knowledge Graph only	20%	55%

Analysis

The ablation results demonstrate that the redesigned Knowledge Graph contributes substantially more useful clinical information than in the baseline implementation.

In the baseline system, removing the Knowledge Graph produced virtually no change in overall diagnostic accuracy, indicating that graph retrieval contributed little to the final decision. By contrast, in the proposed framework, removing the Knowledge Graph reduced **Top-1 Accuracy** from **83%** to **73%**, indicating that graph-derived evidence became a meaningful contributor to the diagnostic reasoning process.

Furthermore, the standalone performance of the Knowledge Graph increased from **20%** to **55%**, demonstrating that the redesigned graph retrieval strategy substantially improved the clinical relevance of graph-derived evidence.

This improvement cannot be attributed solely to increased graph connectivity. Rather, it results from integrating clinically meaningful context into graph retrieval, reducing the influence of chronic and clinically irrelevant diseases while promoting evidence that is more compatible with the patient's presentation. This observation is consistent with recent graph-based biomedical retrieval approaches that emphasize structured graph reasoning instead of relying exclusively on symptom associations [2].

Qualitative Comparison of Knowledge Graph Retrieval

Table 6.3 illustrates the qualitative difference between the baseline Knowledge Graph retrieval and the proposed context-aware graph retrieval.

Table 6.3. Qualitative comparison of Knowledge Graph retrieval outputs.

Rank	Dense Retrieval	BM25 Retrieval	Baseline Knowledge Graph	Proposed Context-Aware Knowledge Graph
1	Acute ischemic stroke in the posterior circulation	Management of acute vertigo and cerebellar stroke	Spinocerebellar Ataxia Type 6	Acute Brainstem Ischemic Stroke
2	Vertebrobasilar insufficiency and infarction	Brainstem infarction presenting with isolated vertigo	Hereditary Episodic Ataxia	Posterior Circulation Infarction
3	Cerebellar nodulus infarction	Vestibular neuritis vs. stroke	Ataxia Telangiectasia	Vertebrobasilar Ischemia
4	Vestibular migraine	Benign paroxysmal positional vertigo	Cerebellar Ataxia with Vertigo	Vestibular Neuritis

The baseline graph primarily ranked diseases according to symptom overlap. Consequently, hereditary and chronic neurological disorders, including **Spinocerebellar Ataxia**, **Hereditary Episodic Ataxia**, and **Ataxia Telangiectasia**, frequently appeared among the highest-ranked candidates because of their strong statistical association with vertigo-related symptoms.

In contrast, the proposed graph retrieval strategy incorporates clinical context during evidence ranking. Acute posterior-circulation entities, including **Brainstem Ischemic Stroke**, **Posterior Circulation Infarction**, and **Vertebrobasilar Ischemia**, are promoted when their clinical characteristics better match the patient's presentation.

Importantly, acute non-vascular differential diagnoses, such as **Vestibular Neuritis**, remain within the candidate list when supported by the available evidence. This behaviour indicates that the proposed framework does not enforce a predetermined stroke diagnosis. Instead, it improves the clinical compatibility of graph-derived evidence while preserving appropriate differential diagnoses.

The qualitative comparison therefore demonstrates that the redesigned Knowledge Graph functions as a context-aware evidence source rather than a symptom-frequency ranking mechanism. This behavior aligns with recent graph-based retrieval methodologies in medical RAG systems while extending them through candidate-oriented evidence evaluation and structured integration within the proposed diagnostic framework [2], [7]

6.5 Complex Case Study: Context-Aware Diagnostic Improvement

To further illustrate the practical impact of the proposed framework, the same posterior-circulation case analyzed in Chapter 3 (acute vertigo, ataxia, and dysarthria) was re-evaluated using the enhanced system. Table 6.3 summarizes the outputs produced by the major retrieval components for this representative case.

Analysis

The qualitative difference between the baseline and the proposed framework is substantial. Unlike the baseline implementation, where Knowledge Graph retrieval primarily returned diseases sharing statistical symptom associations, the proposed framework prioritized clinically compatible acute vascular conditions after incorporating clinical context into the reasoning process.

The retrieved evidence now consistently emphasizes concepts such as **acute ischemic stroke**, **brainstem infarction**, **posterior circulation infarction**, and **vertebrobasilar ischemia**, which are clinically consistent with the patient's presentation. At the same time, plausible non-vascular differential diagnoses remain available when supported by the retrieved evidence, thereby preserving diagnostic completeness.

Furthermore, graph-derived evidence no longer acts as an isolated retrieval component. Instead, it reinforces the evidence obtained through Dense Retrieval and BM25 before candidate-specific verification and Evidence Judge evaluation are performed. This complementary behavior produces a more coherent evidence package and improves the reliability of the final diagnostic recommendation. The proposed retrieval strategy is conceptually aligned with graph-based medical retrieval and candidate verification approaches while extending them through context-aware evidence integration [2], [7].

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6.6 Evaluation of Interpretation Quality and Clinical Safety

To assess the quality of the generated clinical interpretations, the same subset of ten representative clinical cases was independently reviewed by two clinical experts using the evaluation protocol described in Chapter

Table 6.4. Interpretation-quality and clinical-safety evaluation.

Evaluation Metric	Baseline System	Proposed Framework
Mean Interpretation Score (1–5)	3.8	4.4
Citation Consistency (0–1)	0.7	0.9
Detected Unsupported Interpretations	2	0

Analysis

The proposed framework achieved higher interpretation quality across all evaluated measures. The average interpretation score increased from **3.8** to **4.4**, indicating that the generated explanations were perceived as clearer, more coherent, and better supported by the retrieved biomedical evidence.

Citation consistency also improved substantially, suggesting stronger agreement between the generated interpretation and the supporting PubMed literature. The experts further noted that the explanations appeared more convincing because multiple evidence sources converged toward the same clinical conclusion rather than presenting partially inconsistent information.

No unsupported interpretations were identified within the evaluated sample. Although this result is limited to the current evaluation dataset, it suggests that combining evidence grounding, candidate-specific verification, and structured evidence evaluation contributes to safer evidence-supported clinical explanations. This observation is consistent with recent work on grounded medical generation and claim-level evidence verification [6], [8], [11].

6.7 Computational Performance

In addition to diagnostic accuracy, the computational impact of the proposed framework was evaluated to determine whether the additional reasoning stages introduced unacceptable execution overhead.

Table 6.5. Computational performance comparison.

Metric	Baseline System	Proposed Framework
Mean Response Time	3.8 s	4.1 s

Analysis

The proposed framework increased the average response time by approximately **0.3 seconds**. This additional latency results from the inclusion of context-aware reasoning, candidate-specific Knowledge Graph verification, and evidence evaluation stages introduced by the proposed architecture.

Despite this modest increase in computational cost, the observed improvement in diagnostic accuracy, evidence quality, and clinical interpretability substantially outweighs the additional processing time. The experimental results therefore indicate that the proposed framework maintains practical computational efficiency while providing more reliable evidence-based clinical decision support.

6.8 Discussion of Results

The experimental results demonstrate that the proposed framework consistently outperformed the baseline implementation across both quantitative and qualitative evaluation measures.

The most significant improvements were observed in clinically challenging posterior-circulation cases, where contextual interpretation of symptoms plays a critical role in distinguishing acute cerebrovascular disorders from chronic, hereditary, or benign neurological conditions. Rather than relying exclusively on symptom overlap, the proposed framework integrates clinical context, hybrid evidence retrieval, Knowledge Graph verification, and structured evidence evaluation to produce diagnostically more relevant recommendations.

Importantly, the observed performance gains cannot be attributed solely to increasing the amount of biomedical knowledge or replacing the language model. Instead, they result from redesigning the reasoning architecture that governs how evidence is retrieved, verified, integrated, and evaluated before a final recommendation is produced.

These findings support the central research hypothesis of this thesis: **incorporating explicit clinical context into evidence retrieval and reasoning significantly improves the reliability, explainability, and diagnostic usefulness of Retrieval-Augmented clinical decision-support systems**

6.9 Final Large-Scale Evaluation on 300 Clinical Cases

To evaluate the overall performance of the proposed framework under a larger and more diverse experimental setting, a final evaluation was conducted using **300 curated clinical case vignettes** distributed across **15 stroke-related diagnostic groups**, with **20 cases per group**.

The evaluation was performed using the fully implemented inference pipeline and the existing biomedical knowledge base without rebuilding the retrieval indexes. Each clinical vignette contained an expected diagnostic label together with the diagnostic output generated by the proposed system.

The evaluation employed an internal **AUTO_CHECK** procedure designed to compare the expected diagnostic label with the final diagnostic recommendation produced by the system. The matching process relied on predefined diagnostic labels together with clinically accepted lexical variants.

Each evaluated case was classified into one of two categories:

- **PASS**, indicating that the expected diagnosis (or one of its accepted variants) was successfully identified by the system according to the predefined matching criteria.
- **REVIEW**, indicating that the expected diagnostic terminology could not be verified automatically or that the system returned insufficient evidence requiring manual inspection.

It is important to emphasize that **AUTO_CHECK measures internal diagnostic agreement rather than externally validated clinical diagnostic accuracy**. Consequently, the reported PASS rates should be interpreted as automated agreement scores within the experimental evaluation framework rather than definitive measures of clinical performance

Overall Results

Across the complete evaluation dataset, **179 of the 300 clinical cases** were classified as **PASS**, corresponding to an overall automated agreement of **59.7%**, whereas **121 cases (40.3%)** required manual review.

Performance varied considerably across diagnostic categories. The highest agreement rates were observed for **Ischemic Stroke** and **Posterior Circulation Stroke**, each achieving **95% PASS**, whereas considerably lower agreement was obtained for **Intracerebral Hemorrhage**, **Transient Ischemic Attack (TIA)**, and several hemorrhagic stroke categories.

The detailed group-level results are presented in Table 6.6.

Table 6.6. Group-level results of the final 300-case AUTO_CHECK evaluation.

Diagnostic group	PASS	REVIEW	Total	PASS rate
Stroke (general)	16	4	20	80%
Ischemic stroke	19	1	20	95%
Subarachnoid hemorrhage	12	8	20	60%
Intracerebral hemorrhage	4	16	20	20%
TIA abbreviation	8	12	20	40%
Transient ischemic attack (full)	7	13	20	35%
Hemorrhagic stroke	7	13	20	35%
Vertebrobasilar	16	4	20	80%
Large vessel occlusion	10	10	20	50%
Basilar artery occlusion	12	8	20	60%
Lacunar infarction	11	9	20	55%
Cerebellar infarction	11	9	20	55%
Anterior circulation stroke	12	8	20	60%
Posterior circulation stroke	19	1	20	95%
Brainstem infarction	15	5	20	75%
Overall	179	121	300	59.7%

Interpretation of Results

The large-scale evaluation demonstrates that the proposed framework performs reliably for several ischemic and posterior-circulation stroke presentations while highlighting remaining challenges for hemorrhagic stroke, transient ischemic attack, and terminology-sensitive diagnostic categories.

These findings suggest that the proposed context-aware reasoning architecture is particularly effective when the retrieved biomedical evidence exhibits strong consistency across multiple retrieval components. Conversely, lower agreement rates frequently occurred in categories characterized by overlapping terminology, heterogeneous clinical presentations, or limited retrieval support.

The observed variability also illustrates one of the principal challenges of biomedical Retrieval-Augmented clinical decision support: semantic similarity alone is insufficient to guarantee complete diagnostic agreement, particularly when multiple clinically related diagnostic concepts share overlapping terminology. Similar observations have been reported in recent work on evidence-grounded medical retrieval and contradiction-aware reasoning [6], [11].

Evaluation Limitations

Although the 300-case evaluation provides substantially broader coverage than the earlier pilot experiments, several limitations should be acknowledged.

First, the current evaluation relies on an automated keyword-based agreement procedure. Consequently, clinically related parent or child diagnostic concepts may occasionally be classified as **REVIEW** despite representing medically compatible diagnoses. Conversely, lexical agreement alone does not necessarily establish complete clinical equivalence.

Second, the current evaluation measures internal system agreement rather than independent clinical correctness. Future evaluations should therefore incorporate:

- blinded expert annotation;
- ontology-aware diagnostic matching;
- confidence interval estimation;
- external validation using real-world clinical cases.

These recommendations are consistent with recent research emphasizing evidence verification, explainability, and clinically grounded evaluation for medical RAG systems [8], [11].

Relationship to Earlier Experiments

The earlier experimental studies involving **30 representative clinical cases** remain valuable as pilot experiments used to identify failure modes, evaluate individual retrieval components, and guide successive improvements to the proposed framework.

However, the **300-case evaluation** represents the principal end-to-end evaluation of the completed system and therefore provides the primary quantitative assessment reported in this thesis.

Overall, the final evaluation demonstrates promising performance for multiple stroke categories while simultaneously identifying important areas requiring further refinement before deployment in real clinical environments.

6.10 Chapter Conclusion

This chapter presented a comprehensive experimental evaluation of the proposed context-aware hybrid clinical decision-support framework.

Both the pilot experiments and the final large-scale evaluation demonstrated that incorporating clinical context into biomedical evidence retrieval and evaluation improves the overall reliability of the diagnostic pipeline. In particular, the proposed framework showed substantial improvement in complex posterior-circulation stroke cases and increased the contribution of structured Knowledge Graph evidence to the overall decision-making process.

The evaluation further demonstrated that integrating hybrid retrieval, candidate-specific Knowledge Graph verification, evidence grounding, and structured evidence evaluation produces more consistent diagnostic recommendations and more interpretable evidence-supported clinical explanations.

Although the large-scale evaluation also revealed limitations in several hemorrhagic and terminology-sensitive diagnostic groups, these findings provide valuable guidance for future improvements in ontology-aware retrieval, evidence verification, and clinical validation.

Overall, the experimental results support the central hypothesis of this research: **integrating explicit clinical context with evidence-guided retrieval and structured reasoning enhances the reliability, explainability, and diagnostic usefulness of Retrieval-Augmented clinical decision-support systems for stroke diagnosis.**

Chapter 7: Future Prospects and Conclusion

7.1 Introduction

The previous chapters presented the design, implementation, and experimental evaluation of the proposed context-aware hybrid clinical decision-support framework for stroke diagnosis. The obtained results demonstrated that integrating clinical context, hybrid evidence retrieval, Knowledge Graph verification, and evidence-guided reasoning can significantly improve diagnostic support while maintaining transparency and explainability.

Despite these encouraging results, the proposed framework should be regarded as an evolving research platform rather than a complete clinical decision-support system. Accordingly, this chapter discusses the current limitations of the proposed implementation, outlines several directions for future research, and concludes by summarizing the principal scientific contributions of this thesis.

7.2 Current Limitations

Although the proposed framework demonstrated encouraging experimental performance, several limitations remain that should be addressed before deployment in real clinical environments.

7.2.1 Limited Temporal Clinical Reasoning

The current implementation has only limited capability to reason explicitly about temporal clinical information.

Although the framework detects several acute clinical patterns, it does not yet perform comprehensive temporal reasoning capable of distinguishing clinical scenarios based on symptom duration or temporal evolution. For example, differentiating a transient ischemic attack (TIA) lasting several minutes from an evolving ischemic stroke still requires more sophisticated temporal event modeling.

Future work should incorporate dedicated temporal information extraction methods capable of recognizing expressions such as *"five minutes ago," "two hours earlier,"* or *"progressively worsening over three days"* and integrating these temporal relationships into the evidence evaluation pipeline.

7.2.2 Limited Knowledge Sources

The biomedical knowledge base currently relies primarily on scientific literature retrieved from PubMed.

Although PubMed provides high-quality peer-reviewed biomedical evidence, the current implementation does not yet integrate additional clinical knowledge sources such as evidence-based clinical guidelines, neurology textbooks, stroke management protocols, or multilingual biomedical literature.

Future work should investigate multi-source retrieval strategies capable of combining scientific publications with structured clinical guidelines and other authoritative medical resources.

7.2.3 Absence of Multimodal Clinical Evidence

The current framework operates exclusively on textual clinical descriptions.

In routine clinical practice, stroke diagnosis depends not only on patient history but also on neuroimaging studies (CT and MRI), vascular imaging, laboratory investigations, electrocardiography, and other diagnostic modalities.

Future versions of the proposed framework could integrate multimodal medical foundation models capable of combining textual evidence with radiological and laboratory information to produce more comprehensive clinical decision support.

7.2.4 Limited Clinical Scope

The proposed framework was specifically designed for stroke-oriented clinical decision support.

Although this specialization enabled detailed modeling of stroke-related reasoning, the current implementation cannot yet support broader neurological disorders or general multi-specialty diagnostic reasoning.

Future work may generalize the proposed architecture by extending the Knowledge Graph, retrieval corpus, and evidence evaluation framework to additional neurological and internal medicine domains.

7.2.5 Evaluation Limitations

Although the experimental evaluation demonstrated promising results, several limitations remain.

The evaluation dataset, while substantially larger than the pilot experiments, does not fully represent the diversity and complexity encountered in routine clinical practice. Furthermore, the current evaluation relied primarily on curated clinical cases and automated agreement measures rather than prospective clinical validation.

Future evaluation should incorporate:

- blinded clinician assessment;
- real-world hospital datasets;
- ontology-aware diagnostic matching;
- prospective clinical validation;
- statistical confidence intervals.

7.2.6 Dependence on Retrieved Evidence

The quality of the final diagnostic recommendation remains dependent on the quality of the retrieved biomedical evidence.

Although the proposed framework incorporates hybrid retrieval, candidate-specific Knowledge Graph verification, grounding assessment, and Evidence Judge evaluation, insufficient or conflicting biomedical evidence may still limit diagnostic confidence.

Future work should investigate adaptive retrieval strategies, contradiction-aware evidence fusion, and continuous knowledge-base updating to improve robustness under challenging clinical scenarios [4], [6], [11].

7.2.7 Clinical Deployment

The proposed framework has been evaluated within a research environment and has not yet undergone prospective deployment in routine clinical practice.

Consequently, issues related to clinician workflow integration, usability, electronic health record interoperability, regulatory approval, and patient safety require further investigation before clinical adoption.

7.3 Future Research Roadmap

Building upon the findings of this thesis, several research directions can further improve the proposed context-aware clinical decision-support framework. These directions are organized according to short-, medium-, and long-term development objectives.

7.3.1 Short-Term Objectives (6–12 Months)

Advanced Temporal Clinical Reasoning

Future work should extend the current clinical-context analysis by incorporating a dedicated temporal reasoning module capable of extracting symptom duration, onset time, progression, and temporal relationships from free-text clinical descriptions.

This extension would improve discrimination between conditions with similar symptoms but different temporal characteristics, such as transient ischemic attack (TIA) and acute ischemic stroke.

Expansion of the Evaluation Dataset

The current evaluation should be extended to include substantially larger collections of curated clinical cases covering a broader spectrum of stroke presentations and differential diagnoses.

Future evaluation should also involve multiple neurologists to establish expert-validated reference diagnoses and reduce dependence on automated agreement measures.

Automatic Knowledge Base Updating

An automated knowledge maintenance pipeline could periodically retrieve newly published PubMed articles, perform preprocessing and semantic indexing, and continuously update both retrieval indexes and the Knowledge Graph without requiring manual intervention.

7.3.2 Medium-Term Objectives (1–3 Years)

Transition toward Multi-Agent Clinical Reasoning

Future versions of the framework may adopt a distributed agentic architecture in which specialized reasoning agents collaborate throughout the diagnostic process.

Possible agents include:

- Clinical Context Agent
- Retrieval Agent
- Candidate Verification Agent
- Evidence Evaluation Agent
- Explanation Agent

Such an architecture would improve modularity, maintainability, and scalability while remaining consistent with recent developments in medical agentic AI [1], [10].

Expansion to Additional Neurological Disorders

The current implementation focuses exclusively on stroke-related diagnosis.

Future work should expand the retrieval corpus, Knowledge Graph, and evidence evaluation framework to include additional neurological disorders such as seizures, multiple sclerosis, brain tumors, peripheral vestibular disorders, and neurodegenerative diseases.

Learning from Clinical Feedback

Future versions could incorporate clinician feedback into the evidence evaluation process, allowing continuous refinement of retrieval strategies, candidate ranking, and evidence weighting while preserving explainability.

7.3.3 Long-Term Objectives (3–5 Years)

Multimodal Clinical Decision Support

One of the most promising research directions is the integration of multimodal medical foundation models capable of combining textual clinical descriptions with CT, MRI, vascular imaging, laboratory investigations, and other diagnostic modalities.

Such integration would enable the framework to perform evidence-guided reasoning across heterogeneous clinical data sources rather than relying exclusively on textual biomedical evidence.

Prospective Clinical Validation

Before deployment in routine healthcare environments, the proposed framework should undergo prospective clinical validation using anonymized hospital datasets and blinded clinician evaluation.

Future studies should compare the framework against physician decision making while evaluating diagnostic accuracy, interpretability, clinical usability, and patient safety.

7.3.4 Explainability and Evidence Verification

Future work should extend the current explainability framework toward complete claim-level evidence verification inspired by SourceCheckup [8].

Potential extensions include:

- independent claim-level support assessment;
- contradiction detection;
- evidence aggregation across multiple biomedical sources;
- clinician-validated support annotations;
- ontology-aware evidence tracing.

These additions would strengthen the transparency of the diagnostic process by ensuring that every generated clinical statement remains explicitly linked to supporting biomedical evidence.

7.4 General Conclusion

This thesis presented the design, implementation, and evaluation of a context-aware hybrid Retrieval-Augmented clinical decision-support framework for stroke diagnosis.

The research began with the objective of developing an evidence-based medical decision-support system capable of assisting clinicians in the diagnosis of acute stroke. During the development process, systematic experimentation revealed that conventional graph-based reasoning was insufficient for reliable clinical decision support because statistical symptom associations alone could not adequately distinguish acute cerebrovascular disorders from chronic or clinically unrelated neurological conditions.

Rather than treating this limitation as an implementation issue, this research addressed it as a fundamental reasoning problem. The proposed solution introduced a context-aware evidence-guided architecture integrating hybrid retrieval, candidate-specific Knowledge Graph verification, evidence grounding, structured evidence evaluation, and claim-level explainability into a unified clinical decision-support framework.

Experimental evaluation demonstrated consistent improvements across several challenging stroke categories, particularly posterior-circulation stroke, while also increasing the contribution of graph-derived evidence to the overall reasoning process. The final large-scale evaluation on **300 curated clinical cases** achieved **179 AUTO_CHECK PASS results (59.7%)**, demonstrating promising performance while simultaneously identifying important limitations requiring further refinement and external clinical validation.

The proposed framework also incorporates explainability mechanisms inspired by recent evidence-verification research, ensuring that diagnostic recommendations remain traceable to supporting biomedical literature rather than relying solely on generated text [8].

Overall, this research demonstrates that integrating explicit clinical context with evidence-guided retrieval and structured reasoning can substantially improve the reliability, interpretability, and transparency of Retrieval-Augmented clinical decision-support systems.

Although further work is required before clinical deployment, the proposed architecture provides a reproducible foundation for future research on trustworthy medical artificial intelligence.

7.5 Summary of Contributions

The principal contributions of this thesis can be summarized as follows:

- **Development of a context-aware hybrid clinical decision-support framework** integrating dense retrieval, lexical retrieval, Knowledge Graph reasoning, biomedical reranking, evidence evaluation, and explainability into a unified Retrieval-Augmented architecture for stroke diagnosis.
- **Identification of a critical limitation in conventional graph-based medical retrieval**, demonstrating that symptom-frequency relationships alone are insufficient for clinically reliable diagnostic reasoning.
- **Design and implementation of a context-aware Knowledge Graph reasoning strategy** capable of incorporating clinically meaningful contextual information into graph retrieval and candidate prioritization.
- **Development of a structured evidence evaluation framework** combining hybrid retrieval, candidate-specific Knowledge Graph verification, grounding assessment, and Evidence Judge evaluation before producing the final diagnostic recommendation.
- **Integration of claim-level explainability and evidence traceability**, enabling every diagnostic recommendation to remain linked to supporting biomedical evidence and corresponding PubMed references.
- **Comprehensive experimental evaluation**, including pilot studies, component-level analysis, qualitative case studies, and large-scale evaluation using 300 curated clinical cases.
- **A reproducible architecture for trustworthy medical Retrieval-Augmented Generation**, providing a foundation for future research on explainable, evidence-grounded, and clinically reliable medical decision-support systems.

APPENDICES

The appendices preserve the final access-layer and configuration files supplied for this thesis. Core retrieval and reasoning modules are described in Chapters 4 and 5 and remain available in the accompanying project repository.

Appendix A: FastAPI Service Entry Point (api.py)

```
# -*- coding: utf-8 -*-
"""FastAPI entry point for the MedRAG agentic clinical decision-support API."""
from __future__ import annotations

from typing import Any, Dict

import yaml
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel, Field

from app.rag.pipeline import get_answer


API_VERSION = "v7.15-agentic-clinical-safety-explainability"

app = FastAPI(
    title="MedRAG Agentic API",
    description="Evidence-Based Agentic RAG API for stroke-oriented clinical decision support",
    version=API_VERSION,
)


class DiagnoseRequest(BaseModel):
    case_text: str = Field(..., min_length=3, description="Clinical case narrative")
    top_k: int = Field(default=30, ge=5, le=100)
    debug: bool = True
    return_explainability: bool = True


def _load_config() -> Dict[str, Any]:
    try:
        with open("config.yaml", "r", encoding="utf-8") as f:
            return yaml.safe_load(f) or {}
    except Exception:
        return {}


def _compact_result(result: Dict[str, Any]) -> Dict[str, Any]:
    """Return a stable compact response without deleting clinical safety fields."""
    keep = {
        "answer",
        "confidence",
        "top_score",
        "pipeline_version",
        "used_fallback",
        "agentic_mode",
        "grounding",
        "selected_candidate",
        "safety_flags",
        "legacy_rules_used",
        "legacy_resolver_used",
    }
    compact = {key: value for key, value in result.items() if key in keep}
    compact["explainability_included"] = False
    return compact


@app.get("/")
def root_health_check() -> Dict[str, Any]:
    return health_check()


@app.get("/health")
def health_check() -> Dict[str, Any]:
```

```
    cfg = _load_config()
    agentic_cfg = cfg.get("agentic_mode", {}) or {}
    xai_cfg = cfg.get("explainability", {}) or {}
    kg_cfg = cfg.get("knowledge_graph", {}) or {}
    return {
        "status": "ok",
        "service": "MedRAG Agentic API",
        "api_version": API_VERSION,
        "endpoint": "/diagnose",
        "agentic_mode_enabled": bool(agentic_cfg.get("enabled", True)),
        "sourcecheckup_enabled": bool(xai_cfg.get("enabled", True)),
        "explainability_methodology": xai_cfg.get("methodology", "sourcecheckup"),
        "knowledge_graph_enabled": bool(kg_cfg.get("enabled", False)),
        "clinical_safety_version": (cfg.get("clinical_safety", {}) or {}).get("version", "v7.15-
negation-temporal-kg-quality"),
        "negation_aware": True,
        "tia_requires_explicit_resolution": True,
        "subtype_requires_neuroimaging_confirmation": True,
    }


@app.post("/diagnose")
def diagnose(req: DiagnoseRequest) -> Dict[str, Any]:
    try:
        result = get_answer(
            case_text=req.case_text.strip(),
            top_k=int(req.top_k),
            debug=bool(req.debug),
        )
        if not isinstance(result, dict):
            raise RuntimeError("The MedRAG pipeline returned a non-dictionary response.")

        if not req.return_explainability:
            return _compact_result(result)

        result = dict(result)
        result["explainability_included"] = isinstance(result.get("explanation"), dict)
        result.setdefault("explanation", {})
        return result

    except HTTPException:
        raise
    except ValueError as exc:
        raise HTTPException(status_code=422, detail=str(exc)) from exc
    except Exception as exc:
        # Avoid exposing stack traces, credentials, model paths, or configuration.
        raise HTTPException(
            status_code=500,
            detail=f"MedRAG processing failed: {type(exc).__name__}",
        ) from exc
```

Appendix B: Streamlit Explainability Interface (app.py)

```
# -*- coding: utf-8 -*-
"""Streamlit UI for candidate-level and claim-level MedRAG explainability."""
from __future__ import annotations

import math
from typing import Any, Dict, Iterable, List, Mapping

import pandas as pd
import requests
import streamlit as st


st.set_page_config(
    page_title="MedRAG — Agentic Explainable Stroke CDSS",
    page_icon="🩺",
    layout="wide",
)

STATUS_ICONS = {
    "supported": "☑",
    "partially_supported": "◍",
    "not_verifiable": "◯",
    "unsupported": "✖",
    "contradicted": "⛔",
    "weak": "⚠",
    "rejected": "✖",
}


def safe_float(value: Any, default: float | None = None) -> float | None:
    try:
        if value is None:
            return default
        number = float(value)
        if math.isnan(number) or math.isinf(number):
            return default
        return number
    except Exception:
        return default


def fmt_num(value: Any, ndigits: int = 3) -> str:
    number = safe_float(value)
    return "—" if number is None else f"{number:.{ndigits}f}"


def as_list(value: Any) -> List[Any]:
    if value is None:
        return []
    if isinstance(value, list):
        return value
    if isinstance(value, tuple):
        return list(value)
    return [value]


def as_dict(value: Any) -> Dict[str, Any]:
    return dict(value) if isinstance(value, Mapping) else {}


def get_nested(data: Any, path: Iterable[str], default: Any = None) -> Any:
    current = data
    for key in path:
```

```
        if not isinstance(current, Mapping):
            return default
        current = current.get(key, default)
    return current


def get_explanation(data: Mapping[str, Any]) -> Dict[str, Any]:
    return as_dict(data.get("explanation"))


def normalize_for_dataframe(rows: Any) -> pd.DataFrame:
    normalized: List[Dict[str, Any]] = []
    for row in as_list(rows):
        if isinstance(row, Mapping):
            normalized.append(dict(row))
        else:
            normalized.append({"value": row})
    return pd.DataFrame(normalized)


def source_is_kg(source: Mapping[str, Any]) -> bool:
    return bool(
        source.get("from_kg")
        or source.get("graph_score") is not None
        or source.get("kg_paths")
        or "kg" in str(source.get("retrieval_channel") or "").lower()
    )


def get_candidate_rows(data: Mapping[str, Any]) -> List[Dict[str, Any]]:
    explanation = get_explanation(data)
    rows = explanation.get("candidate_comparison") or data.get("evidence_judgments") or []
    normalized = [dict(row) for row in as_list(rows) if isinstance(row, Mapping)]
    normalized.sort(
        key=lambda row: safe_float(row.get("final_support_score", row.get("support_score")), 0.0) or
0.0,
        reverse=True,
    )
    return normalized


def get_agent_steps(data: Mapping[str, Any]) -> List[Any]:
    explanation = get_explanation(data)
    candidates = [
        data.get("agent_steps"),
        get_nested(explanation, ["developer_debug_summary", "debug_info", "agent_steps"]),
        get_nested(explanation, ["developer_debug_summary", "agent_steps"]),
        get_nested(data, ["debug_info", "agent_steps"]),
    ]
    for value in candidates:
        if isinstance(value, list) and value:
            return value
    return []


def render_summary_cards(data: Mapping[str, Any]) -> None:
    explanation = get_explanation(data)
    metrics = as_dict(explanation.get("metrics"))
    rows = get_candidate_rows(data)
    evidence = as_list(explanation.get("supporting_evidence") or data.get("sources"))

    final_info = as_dict(explanation.get("final_diagnosis"))
    answer = data.get("answer") or explanation.get("final_answer") or "—"
    display_answer = final_info.get("display_label") or answer
    confidence = data.get("confidence") or explanation.get("confidence_level") or "—"
    support_score = explanation.get("support_score")

    st.subheader("☑ النتيجة النهائية")
```

```
    st.markdown(f"## {display_answer}")
    if final_info.get("provisional"):
        st.warning("هذا تصنيف مبدئي مدعوم بالأدلة، وليس تأكيدًا لنوع السكتة .يلزم تصوير دماغ عاجل لاستبعاد النزف
وتأكيد النوع.")

    cols = st.columns(7)
    cols[0].metric("الثقة", str(confidence))
    cols[1].metric("Final Support", fmt_num(support_score))
    cols[2].metric("المرشحون", len(rows))
    cols[3].metric("الأدلة", len(evidence))
    cols[4].metric("Claim Support", fmt_num(metrics.get("claim_support_rate")))
    cols[5].metric("Citation Coverage", fmt_num(metrics.get("citation_coverage")))
    cols[6].metric("Fallback", str(data.get("used_fallback", False)))

    if data.get("used_fallback") or explanation.get("limited_explainability"):
        st.warning("تم استخدام Legacy Fallback محدودًا؛ لذلك قد يكون تفسير قرار المرشحين .")


def render_why_selected(data: Mapping[str, Any]) -> None:
    explanation = get_explanation(data)
    narrative = explanation.get("decision_narrative")
    why = as_dict(explanation.get("why_selected"))

    st.subheader("لماذا اختار النظام هذا التشخيص؟ 🤔")
    if narrative:
        st.info(str(narrative))
    else:
        st.warning("لا يوجد Decision Narrative في الاستجابة الحالية.")

    main_reasons = as_list(why.get("main_reasons"))
    if main_reasons:
        st.markdown("### الأسباب الرئيسية")
        for reason in main_reasons:
            st.markdown(f"- {reason}")

    col1, col2 = st.columns(2)
    with col1:
        st.markdown("### أقوى الإشارات السريرية")
        signals = as_list(why.get("strongest_clinical_signals"))
        if signals:
            for signal in signals:
                st.markdown(f"- {signal}")
        else:
            st.caption("لا توجد إشارات سريرية موثقة في هذا الخرج.")

        st.markdown("### Grounding")
        st.write(why.get("grounding_summary") or explanation.get("grounding_explanation") or "—")

    with col2:
        st.markdown("### Knowledge Graph")
        st.write(why.get("kg_support_summary") or "—")

        st.markdown("### أفضلية السكور")
        advantages = normalize_for_dataframe(why.get("score_advantages"))
        if not advantages.empty:
            st.dataframe(advantages, use_container_width=True, hide_index=True)
        else:
            st.caption("لا توجد بدائل للمقارنة.")

    st.markdown("### أقوى المقالات الداعمة")
    sources = as_list(why.get("strongest_supporting_sources"))
    if sources:
        for source in sources:
            if not isinstance(source, Mapping):
                continue
            title = source.get("title") or "مصدر بلا عنوان"
            pmid = source.get("pmid") or "—"
```

```
            st.markdown(f"- **{title}** — PMID: `{pmid}` — MedCPT:
`{fmt_num(source.get('medcpt_score'))}`")
    else:
        st.caption("لم تصل مقالات داعمة مرتبطة بالمرشح المختار.")

    with st.expander("بيانات القرار المنظمة"):
        st.json(why)


def render_candidates(data: Mapping[str, Any]) -> None:
    st.subheader("🏁 التشخيصات المرشحة ومكونات القرار")
    rows = get_candidate_rows(data)
    if not rows:
        st.warning("لا توجد Candidate Comparison في الاستجابة.")
        return

    table_rows: List[Dict[str, Any]] = []
    for index, row in enumerate(rows, start=1):
        breakdown = as_dict(row.get("score_breakdown"))
        selected = bool(row.get("selected"))
        decision = str(row.get("decision") or "unknown")
        icon = "☑" if selected else STATUS_ICONS.get(decision, "◯")
        table_rows.append({
            "Rank": row.get("rank", index),
            "Status": icon,
            "Candidate": row.get("candidate_label") or row.get("candidate") or
row.get("candidate_key"),
            "Decision": decision,
            "Final score": safe_float(row.get("final_support_score", row.get("support_score")), 0.0),
            "Text": safe_float(breakdown.get("textual_support", row.get("textual_support")), 0.0),
            "KG": safe_float(breakdown.get("kg_support", row.get("kg_support")), 0.0),
            "Grounding": safe_float(breakdown.get("grounding_support", row.get("grounding_support")),
0.0),
            "Reranker": safe_float(breakdown.get("rerank_strength", row.get("rerank_strength")), 0.0),
            "Diversity": safe_float(breakdown.get("source_diversity", row.get("source_diversity")),
0.0),
            "Intent": safe_float(breakdown.get("intent_compatibility",
row.get("intent_compatibility")), 0.0),
            "Penalty": safe_float(breakdown.get("contradiction_penalty",
row.get("contradiction_penalty")), 0.0),
        })
    st.dataframe(pd.DataFrame(table_rows), use_container_width=True, hide_index=True)

    for row in rows:
        title = row.get("candidate_label") or row.get("candidate") or "Candidate"
        selected = bool(row.get("selected"))
        icon = "☑" if selected else STATUS_ICONS.get(str(row.get("decision")), "◯")
        with st.expander(f"{icon} {title} — score {fmt_num(row.get('final_support_score',
row.get('support_score')))}", expanded=selected):
            reasons = row.get("selection_reasons") if selected else row.get("why_not_selected")
            heading = "لماذا تم اختياره؟" if selected else "لماذا لم يتم اختياره؟"
            st.markdown(f"### {heading}")
            for reason in as_list(reasons):
                st.markdown(f"- {reason}")
            st.markdown("### Score breakdown")
            st.json(as_dict(row.get("score_breakdown")))


def render_evidence(data: Mapping[str, Any]) -> None:
    st.subheader("📑 الأدلة والمقالات الداعمة")
    explanation = get_explanation(data)
    evidence = as_list(explanation.get("supporting_evidence") or explanation.get("evidence_trace"))
    if not evidence:
        st.info("لا توجد أدلة داعمة مرتبطة بالمرشح المختار.")
        return

    for index, item in enumerate(evidence, start=1):
```

```
        if not isinstance(item, Mapping):
            continue
        title = item.get("title") or "مصدر بلا عنوان"
        pmid = item.get("pmid") or "—"
        status = str(item.get("support_status") or "not_verifiable")
        icon = STATUS_ICONS.get(status, "◯")
        with st.expander(f"{index}. {icon} {title} — PMID {pmid}", expanded=(index <= 2)):
            source_url = item.get("source_url") or item.get("source_name")
            if source_url:
                st.markdown(f"[فتح المصدر]({source_url})")

            st.markdown("**المقتطف الداعم:**")
            st.write(item.get("exact_supporting_span") or item.get("snippet") or
item.get("evidence_text") or "—")

            st.markdown("**لماذا هذا الدليل مهم؟**")
            st.write(item.get("why_relevant") or "لم يتم التحقق من رابط Claim محدد لهذا المصدر.")

            claim_ids = as_list(item.get("supported_claim_ids"))
            st.write("**Claims المدعومة:**", ", ".join(map(str, claim_ids)) if claim_ids else "—")

            scores = as_dict(item.get("scores")) or {
                "dense_score": item.get("dense_score"),
                "bm25_score": item.get("bm25_score"),
                "hybrid_score": item.get("hybrid_score"),
                "medcpt_score": item.get("medcpt_score"),
                "graph_score": item.get("graph_score"),
                "grounding_strength": item.get("grounding_strength"),
            }
            score_cols = st.columns(6)
            labels = ["Dense", "BM25", "Hybrid", "MedCPT", "Graph", "Grounding"]
            keys = ["dense_score", "bm25_score", "hybrid_score", "medcpt_score", "graph_score",
"grounding_strength"]
            for col, label, key in zip(score_cols, labels, keys):
                col.metric(label, fmt_num(scores.get(key)))

            st.caption(
                f"Channel: {item.get('retrieval_channel') or '—'} | KG: {source_is_kg(item)} | Status:
{status}"
            )


def render_claims(data: Mapping[str, Any]) -> None:
    st.subheader("🔍 التحقق من الادعاءات — SourceCheckup")
    explanation = get_explanation(data)
    claims = [dict(x) for x in as_list(explanation.get("claims")) if isinstance(x, Mapping)]
    if not claims:
        st.warning("لا توجد Claims في الاستجابة.")
        return

    summary_rows = []
    for claim in claims:
        status = str(claim.get("support_status") or "not_verifiable")
        evidence_count = sum(len(as_list(claim.get(key))) for key in (
            "patient_evidence", "document_evidence", "kg_evidence", "contradicting_evidence"
        ))
        summary_rows.append({
            "Status": STATUS_ICONS.get(status, "◯"),
            "Claim": claim.get("claim_text"),
            "Type": claim.get("claim_type"),
            "Support score": safe_float(claim.get("support_score")),
            "Evidence": evidence_count,
        })
    st.dataframe(pd.DataFrame(summary_rows), use_container_width=True, hide_index=True)

    for claim in claims:
        status = str(claim.get("support_status") or "not_verifiable")
```

```
        icon = STATUS_ICONS.get(status, "◯")
        with st.expander(f"{icon} {claim.get('claim_id')} — {claim.get('claim_text')}"):
            st.write("**نوع الادعاء:**", claim.get("claim_type"))
            st.write("**حالة الدعم:**", status)
            st.write("**سبب التحقق:**", claim.get("verification_reason") or "—")
            st.write("**طريقة التحقق:**", claim.get("verification_method") or "—")

            sections = [
                ("Patient evidence", "patient_evidence"),
                ("Document evidence", "document_evidence"),
                ("KG evidence", "kg_evidence"),
                ("Contradicting evidence", "contradicting_evidence"),
                ("Missing information", "missing_information"),
            ]
            for label, key in sections:
                values = as_list(claim.get(key))
                if values:
                    st.markdown(f"### {label}")
                    st.json(values)


def render_alternatives(data: Mapping[str, Any]) -> None:
    st.subheader("⦸ لماذا لم يتم اختيار البدائل؟")
    explanation = get_explanation(data)
    alternatives = [dict(x) for x in as_list(explanation.get("alternative_diagnoses")) if isinstance(x,
Mapping)]
    if not alternatives:
        st.info("لا توجد تشخيصات بديلة متاحة للشرح.")
        return

    for alt in alternatives:
        title = alt.get("candidate") or alt.get("candidate_label") or alt.get("candidate_key")
        decision = str(alt.get("decision") or "unknown")
        icon = STATUS_ICONS.get(decision, "◯")
        with st.expander(f"{icon} {title} — score {fmt_num(alt.get('final_support_score',
alt.get('support_score')))}", expanded=True):
            st.metric("الفرق عن التشخيص المختار", fmt_num(alt.get("score_difference_from_selected")))
            for reason in as_list(alt.get("why_not_selected") or alt.get("reason")):
                st.markdown(f"- {reason}")
            missing = as_list(alt.get("missing_support"))
            if missing:
                st.markdown("**الدعم الناقص:**")
                for item in missing:
                    st.markdown(f"- {item}")
            conflicts = as_list(alt.get("conflicting_evidence"))
            if conflicts:
                st.markdown("**الأدلة المتعارضة:**")
                st.json(conflicts)


def _path_text(path: Mapping[str, Any]) -> str:
    nodes = as_list(path.get("nodes"))
    relationships = as_list(path.get("relationships"))
    labels: List[str] = []
    for node in nodes:
        if isinstance(node, Mapping):
            labels.append(str(node.get("name") or node.get("label") or node.get("value") or
node.get("type") or "Node"))
        else:
            labels.append(str(node))
    if not labels:
        return str(path.get("path_id") or "KG path")
    parts: List[str] = [labels[0]]
    for index, relationship in enumerate(relationships):
        parts.append(f"→ {relationship} →")
        if index + 1 < len(labels):
            parts.append(labels[index + 1])
```

```
    return " ".join(parts)


def render_kg(data: Mapping[str, Any]) -> None:
    st.subheader("☸ Knowledge Graph Explanation")
    paths = as_list(get_explanation(data).get("kg_explanation_paths"))
    if not paths:
        st.info("لم يرجع Neo4j مسارات KG قابلة للتدقيق لهذه الحالة.")
        return
    for index, path in enumerate(paths, start=1):
        if not isinstance(path, Mapping):
            continue
        st.markdown(f"### Path {index}")
        st.code(_path_text(path), language=None)
        st.caption(f"Verified: {path.get('verified', '—')} | Source: {path.get('source', '—')}")
        with st.expander("Raw KG path"):
            st.json(path)


def render_uncertainty(data: Mapping[str, Any]) -> None:
    st.subheader("❓ عدم اليقين والمعلومات الناقصة")
    explanation = get_explanation(data)
    uncertainty = as_dict(explanation.get("uncertainty"))
    if not uncertainty:
        uncertainty = {
            "level": explanation.get("confidence_level"),
            "reasons": explanation.get("uncertainty_reasons"),
            "missing_clinical_information": explanation.get("missing_clinical_information"),
        }

    st.metric("Uncertainty / confidence level", str(uncertainty.get("level") or "—"))
    for reason in as_list(uncertainty.get("reasons")):
        st.warning(str(reason))

    missing = normalize_for_dataframe(uncertainty.get("missing_clinical_information"))
    if not missing.empty:
        st.markdown("### المعلومات السريرية غير الموثقة")
        st.dataframe(missing, use_container_width=True, hide_index=True)

    col1, col2 = st.columns(2)
    with col1:
        st.markdown("### Unsupported Claims")
        values = as_list(uncertainty.get("unsupported_claims"))
        if values:
            st.json(values)
        else:
            st.caption("لا يوجد.")
    with col2:
        st.markdown("### Not Verifiable Claims")
        values = as_list(uncertainty.get("not_verifiable_claims"))
        if values:
            st.json(values)
        else:
            st.caption("لا يوجد.")

    st.markdown("### Medical safety notes")
    for note in as_list(explanation.get("medical_safety_notes")):
        st.warning(str(note))


def render_agent_trace(data: Mapping[str, Any]) -> None:
    st.subheader("🕵 حركة الـAgent – للمطور والمناقشة")
    steps = get_agent_steps(data)
    if not steps:
        st.info("لا توجد Agent Steps. شغّل debug=True في الاستجابة.")
        return
    for index, step in enumerate(steps, start=1):
```

```
        name = step.get("step", f"step_{index}") if isinstance(step, Mapping) else f"step_{index}"
        with st.expander(f"{index}. {name}", expanded=(index <= 3)):
            if isinstance(step, Mapping):
                st.json(step)
            else:
                st.write(step)


def render_raw_json(data: Mapping[str, Any]) -> None:
    st.subheader("📄 Raw JSON")
    st.json(data)


st.title("🩺 MedRAG — Agentic Explainable Stroke CDSS")
st.write(
    "يعرض النظام التشخيص المختار، مكونات قرار Evidence Judge، المقالات الداعمة، "
    "سبب استبعاد البدائل، والتحقق من كل Claim وفق SourceCheckup."
)

with st.sidebar:
    st.subheader("الإعدادات")
    api_url = st.text_input("API URL", value="http://localhost:8000/diagnose")
    top_k = st.slider("عدد الأدلة top_k", min_value=5, max_value=50, value=30, step=5)
    include_explainability = st.checkbox("إرجاع Explainability كاملة", value=True)
    show_raw_by_default = st.checkbox("فتح Raw JSON افتراضيًا", value=False)

case = st.text_area(
    "أدخل وصف الحالة السريرية",
    height=220,
    placeholder=(
        "مثال: A 72-year-old man developed sudden right-sided weakness, aphasia, "
        "and slurred speech 45 minutes ago. The deficits are still present."
    ),
)

col1, col2 = st.columns(2)
with col1:
    submit = st.button("تحليل الحالة واسترجاع الأدلة", use_container_width=True, type="primary")
with col2:
    if st.button("مسح", use_container_width=True):
        st.rerun()

if submit:
    if not case.strip():
        st.warning("رجاءً أدخل وصف الحالة السريرية.")
    else:
        with st.spinner("جارٍ تشغيل MedRAG Agent وتحليل الأدلة…"):
            try:
                payload = {
                    "case_text": case.strip(),
                    "top_k": int(top_k),
                    "debug": True,
                    "return_explainability": bool(include_explainability),
                }
                response = requests.post(api_url, json=payload, timeout=900)
                response.raise_for_status()
                data = response.json()

                render_summary_cards(data)
                st.divider()

                if not include_explainability:
                    st.info("تم طلب الاستجابة المختصرة. فعّل Explainability من القائمة الجانبية لرؤية تفاصيل القرار.")
                    with st.expander("الاستجابة المختصرة", expanded=True):
                        st.json(data)
                else:
```

```
        tabs = st.tabs([
            "🤔 لماذا هذا التشخيص؟",
            "🏆 المرشحون",
            "📑 الأدلة والمقالات",
            "🔬 Claims",
            "🚫 البدائل",
            "🕸 KG",
            "❓ Uncertainty",
            "🧭 Agent Trace",
            "📄 Raw JSON",
        ])
        with tabs[0]:
            render_why_selected(data)
        with tabs[1]:
            render_candidates(data)
        with tabs[2]:
            render_evidence(data)
        with tabs[3]:
            render_claims(data)
        with tabs[4]:
            render_alternatives(data)
        with tabs[5]:
            render_kg(data)
        with tabs[6]:
            render_uncertainty(data)
        with tabs[7]:
            render_agent_trace(data)
        with tabs[8]:
            if show_raw_by_default:
                render_raw_json(data)
            else:
                with st.expander("Show full raw JSON"):
                    render_raw_json(data)

except requests.exceptions.HTTPError as exc:
    detail = ""
    try:
        detail = str(response.json().get("detail") or "")
    except Exception:
        pass
    st.error(f"خطأ من الـAPI: {exc}. {detail}")
except requests.exceptions.RequestException as exc:
    st.error(f"تعذّر الاتصال بالـAPI: {exc}")
except ValueError:
    st.error("تعذر تفسير استجابة الخادم كـJSON.")
except Exception as exc:
    st.exception(exc)
```

Appendix C: Runtime Configuration (config.yaml)

```yaml
app:
  project_name: "Med-RAG"
  debug: true

kb:
  chunk_size: 1000          # زيادة حجم القطعة لتحسين السياق
  chunk_overlap: 200        # تداخل مناسب

retrieval:
  top_k_dense: 50
  top_k_bm25: 50
  top_k_merged: 30
  max_docs_for_llm: 8       # تقليل العدد لأن القطع أكبر
  max_context_chars: 6000   # زيادة الحد الأقصى
  use_bm25: true
  w_dense: 0.7
  w_bm25: 0.3
  use_reranker: true
  reranker_model: "ncbi/MedCPT-Cross-Encoder"   # نموذج طبي متخصص

models:
  llm_provider: "hf_api"
  llm_model: "Qwen/Qwen2.5-3B-Instruct"   # استخدم النموذج المناسب
  openai_model: "gpt-4o-mini"
  embeddings: "ncbi/MedCPT-Article-Encoder"      # نموذج تضمين طبي
  use_normalizer: null

generation:
  temperature: 0.0
  max_new_tokens: 512
  repetition_penalty: 1.05
  stop:
    - "<|im_end|>"
    - "</s>"
    - "<|endoftext|>"

pubmed:
  enable: false
  retmax: 8
  years_back: 10
  merge_with_local_kb: true

knowledge_graph:
  enabled: true
  uri: "bolt://localhost:7687"
  user: "neo4j"
  password: "password"
  use_hybrid_search: true
  graph_weight: 0.3

paths:
  bm25_index: "app/index/bm25.joblib"
  data_raw: "data/raw"
  data_processed: "data/processed"
  indices: "data/indices"
clinical_signal:
  enabled: true
  use_transformer: true
  transformer_model: "sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2"
  threshold: 0.42
  max_expansions: 6
  use_scispacy: false

# ------------------------------------------------------------------------------
```

```yaml
# Agentic evidence-based path (explicitly enabled).
# Legacy fallback is retained for backward compatibility. pipeline.py marks its
# explanation as limited because no Evidence Judge decision exists on that path.
# ---------------------------------------------------------------------------
agentic_mode:
  enabled: true
  allow_legacy_fallback_on_error: true
  evidence_support_cap: 3
  min_candidate_support_score: 0.66
  candidate_kg_top_k: 5

# The code reads the plural key `clinical_signals`.  The original singular
# `clinical_signal` block is retained above for backward compatibility.
clinical_signals:
  model_enabled: false
  use_dynamic_query_expansion: true
  model_threshold: 0.62

# ---------------------------------------------------------------------------
# SourceCheckup-inspired, post-decision explainability.
# It verifies claims against existing patient data, retrieved evidence,
# Evidence Judge output, and KG paths.  It never changes the final diagnosis.
# ---------------------------------------------------------------------------
explainability:
  enabled: true
  methodology: "sourcecheckup"
  reference: "https://www.nature.com/articles/s41467-025-58551-6"
  claim_level: true
  kg_aware: true
  post_decision: true
  non_interventional: true
  use_llm_verifier: false
  max_evidence_per_claim: 5
  max_patient_fact_claims: 12
  support_threshold: 0.66
  partial_support_threshold: 0.45
  contradiction_threshold: 0.55
  partial_support_weight: 0.5
  candidate_decision_trace: true
  expose_score_breakdown: true
  expose_alternative_reasons: true
  expose_supporting_evidence: true
  narrative_language: "ar"


# Shared negation-aware clinical context used by intent, signals, judge and KG.
clinical_safety:
  version: "v7.15-negation-temporal-kg-quality"
  require_explicit_resolution_for_tia: true
  require_affirmed_hemorrhage_features_for_recall: true
  require_neuroimaging_for_subtype_confirmation: true
  exclude_nonhuman_preclinical_kg_evidence: true
  min_candidate_kg_alignment: 0.08
```

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